VERSION 1 CAPS

GRADE 10
PHYSICAL SCIENCE

WRITTEN BY VOLUNTEERS

SIYAVULA
TECHNOLOGY-POWERED LEARNING

basic education
Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

SHUTTLEWORTH
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Periodic Table of the Elements

1 IA	2 IIA	3 IIIB	4 IVB	5 VB	6 VIB	7 VIIB	8 VII	9 VII	10 VII	11 IB	12 IIB	13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA	18 0
1 2.2 **H** 1,01																	2 **He** 4,00
3 0.98 **Li** 6,94	4 1.57 **Be** 9,01						No EN **Element** AMU					5 2.04 **B** 10,8	6 2.55 **C** 12,0	7 3.04 **N** 14,0	8 3.44 **O** 16,0	9 3.98 **F** 19,0	10 **Ne** 20,2
11 0.93 **Na** 23,0	12 1.31 **Mg** 24,3											13 1.61 **Al** 27,0	14 1.90 **Si** 28,1	15 2.19 **P** 31,0	16 2.58 **S** 32,1	17 3.16 **Cl** 35,45	18 **Ar** 39,9
19 0.93 **K** 39,1	20 1.00 **Ca** 40,1	21 1.36 **Sc** 45,0	22 1.54 **Ti** 47,9	23 1.63 **V** 50,9	24 1.66 **Cr** 52,0	25 1.55 **Mn** 54,9	26 1.33 **Fe** 55,8	27 1.88 **Co** 58,9	28 1.91 **Ni** 58,7	29 1.90 **Cu** 63,5	30 1.65 **Zn** 65,4	31 1.81 **Ga** 69,7	32 2.01 **Ge** 72,6	33 2.18 **As** 74,9	34 2.55 **Se** 79,0	35 2.96 **Br** 79,9	36 **Kr** 83,8
37 0.82 **Rb** 85,5	38 0.95 **Sr** 87,6	39 1.22 **Y** 88,9	40 1.33 **Zr** 91,2	41 1.60 **Nb** 92,9	42 2.16 **Mo** 95,9	43 1.90 **Tc** (98)	44 2.20 **Ru** 101,1	45 2.28 **Rh** 102,9	46 2.20 **Pd** 106,4	47 1.93 **Ag** 107,9	48 1.69 **Cd** 112,4	49 1.78 **In** 114,8	50 1.96 **Sn** 118,7	51 2.05 **Sb** 121,8	52 2.10 **Te** 127,6	53 2.66 **I** 126,9	54 **Xe** 131,3
55 0.79 **Cs** 132,9	56 0.89 **Ba** 137,3	57-71 **La-Lu** Lanthanides	72 1.30 **Hf** 178,5	73 1.50 **Ta** 180,9	74 2.36 **W** 183,8	75 1.90 **Re** 186,2	76 2.20 **Os** 190,2	77 2.20 **Ir** 192,2	78 2.28 **Pt** 195,1	79 2.54 **Au** 197,0	80 2.00 **Hg** 200,6	81 1.62 **Tl** 204,4	82 2.33 **Pb** 207,2	83 2.02 **Bi** 209,0	84 2.00 **Po** (209)	85 2.20 **At** (210)	86 **Rn** (222)
87 0.7 **Fr** (223)	88 0.9 **Ra** 226,0	89-103 **Ac-Lr** Actinides	104 **Rf** (261)	105 **Db** (262)	106 **Sg** (263)	107 **Bh** (262)	108 **Hs** (265)	109 **Mt** (266)	110 **Ds** (269)	111 **Rg** (272)	112 **Cn** (277)	113 **Uut** (284)	114 **Uuq** (289)	115 **Uup** (288)	116 **Uuh** (293)	117 **Uus** (282)	118 **Uuo** (282)

57 1.10 **La** 138,9	58 1.12 **Ce** 140,1	59 1.13 **Pr** 140,9	60 1.14 **Nd** 144,2	61 **Pm** (145)	62 1.17 **Sm** 150,4	63 **Eu** 152,0	64 1.20 **Gd** 157,3	65 **Tb** 158,9	66 1.22 **Dy** 162,5	67 1.23 **Ho** 164,9	68 1.24 **Er** 167,3	69 1.25 **Tm** 168,9	70 **Yb** 173,0	71 1.27 **Lu** 175,0
89 1.10 **Ac** 227,0	90 1.30 **Th** 232,0	91 1.50 **Pa** 231,0	92 1.38 **U** 238,0	93 1.36 **Np** 237,0	94 1.28 **Pu** (244)	95 1.30 **Am** (243)	96 1.30 **Cm** (247)	97 1.30 **Bk** (247)	98 1.30 **Cf** (251)	99 1.30 **Es** (252)	100 1.30 **Fm** (257)	101 1.30 **Md** (258)	102 1.30 **No** (258)	103 **Lr** (260)

- Transition Metal
- Metal
- Metalloid
- Non-metal
- Noble Gas
- Lanthanide
- Actinide

Everything Science

Grade 10 Physical Science

Version 1 – CAPS

by Siyavula and volunteers

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Everything Science

When we look outside at everything in nature, look around us at everything manufactured or look up at everything in space we cannot but be struck by the incredible diversity and complexity of life; so many things, that look so different, operating in such unique ways. The physical universe really contains incredible complexity.

Yet, what is even more remarkable than this seeming complexity is the fact that things in the physical universe are knowable. We can investigate them, analyse them and understand them. It is this ability to understand the physical universe that allows us to transform elements and make technological progress possible.

If we look back at some of the things that developed over the last century – space travel, advances in medicine, wireless communication (from television to mobile phones) and materials a thousand times stronger than steel – we see they are not the consequence of magic or some inexplicable phenomena. They were all developed through the study and systematic application of the physical sciences. So as we look forward at the 21st century and some of the problems of poverty, disease and pollution that face us, it is partly to the physical sciences we need to turn.

For however great these challenges seem, we know that the physical universe is knowable and that the dedicated study thereof can lead to the most remarkable advances. There can hardly be a more exciting challenge than laying bare the seeming complexity of the physical universe and working with the incredible diversity therein to develop products and services that add real quality to people's lives.

Physical sciences is far more wonderful, exciting and beautiful than magic! It is everywhere. See introductory video by Dr. Mark Horner: ▶ VPsfk at www.everythingscience.co.za

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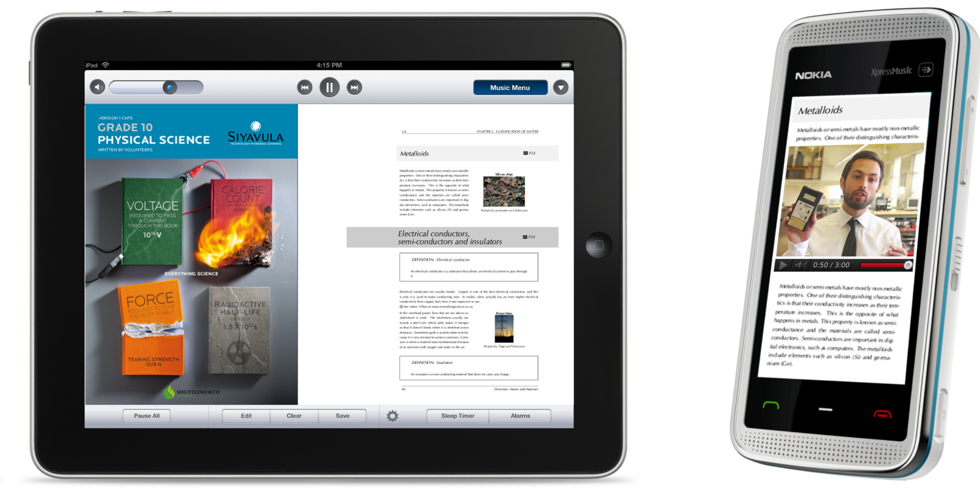

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(V123) Video, simulation or presentation

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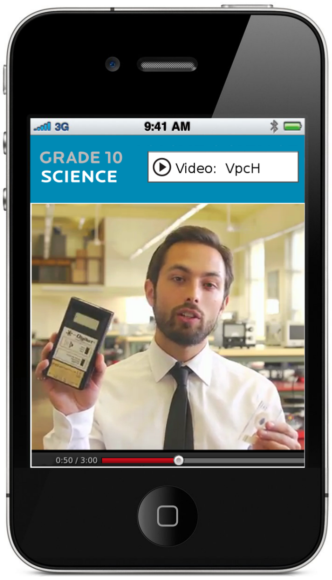

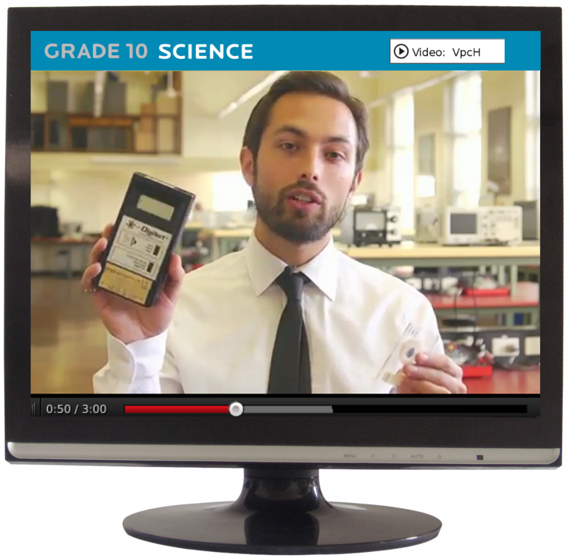

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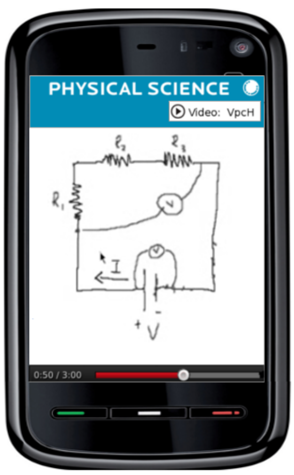

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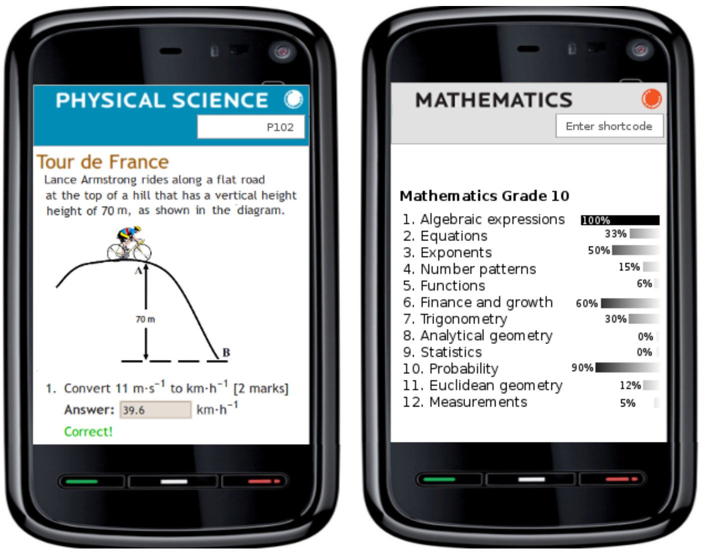

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Contents

Skills for science 1

Introduction ESAA

In the physical sciences there are many skills that you need to learn. These include working with units, basic mathematics skills and laboratory skills. In this chapter we will revise some of these skills that you should know before starting to study physical science. This chapter is intended as a reference guide to assist you in your journey of studying physical science.

See introductory video: (Video: VPrny at www.everythingscience.co.za)

Mathematical skills ESAB

You should be comfortable with scientific notation and how to write scientific notation. You should also be able to easily convert between different units and change the subject of a formula. In addition, concepts such as rate, direct and indirect proportion, fractions and ratios and the use of constants in equations are important.

Rounding off ESAC

Certain numbers may take an infinite amount of paper and ink to write out. Not only is that impossible, but writing numbers out to a high precision (many decimal places) is very inconvenient and rarely gives better answers. For this reason we often estimate the number to a certain number of decimal places.

Rounding off a decimal number to a given number of decimal places is the quickest way to approximate a number. For example, if you wanted to round-off $2,6525272$ to three decimal places then you would first count three places after the decimal. Next you mark this point with a $|$: $2,652|5272$. All numbers to the right of $|$ are ignored after you determine whether the number in the third decimal place must be rounded up or rounded down. You *round up* the final digit (make the digit one more) if the first digit after the $|$ is greater than

or equal to 5 and *round down* (leave the digit alone) otherwise. So, since the first digit after the | is a 5, we must round up the digit in the third decimal place to a 3 and the final answer of $2,6525272$ rounded to three decimal places is $2,653$.
In a calculation that has many steps, it is best to leave the rounding off right until the end. This ensures that your answer is more accurate.

Scientific notation

ESAD

In science one often needs to work with very large or very small numbers. These can be written more easily (and more compactly) in scientific notation, in the general form:

$$N \times 10^n$$

where N is a decimal number between 0 and 10 that is rounded off to a few decimal places. n is known as the *exponent* and is an integer. If $n > 0$ it represents how many times the decimal place in N should be moved to the right. If $n < 0$, then it represents how many times the decimal place in N should be moved to the left. For example $3,24 \times 10^3$ represents $3\ 240$ (the decimal moved three places to the right) and $3,24 \times 10^{-3}$ represents $0,00324$ (the decimal moved three places to the left).
If a number must be converted into scientific notation, we need to work out how many times the number must be multiplied or divided by 10 to make it into a number between 1 and 10 (i.e. the value of n) and what this number between 1 and 10 is (the value of N). We do this by counting the number of decimal places the decimal comma must move.
For example, write the speed of light ($299\ 792\ 458$ m $\cdot$ s^{-1}) in scientific notation, to two decimal places. First, we find where the decimal comma must go for two decimal places (to find N) and then count how many places there are after the decimal comma to determine n.
In this example, the decimal comma must go after the first 2, but since the number after the 9 is 7, $N = 3,00$. $n = 8$ because there are 8 digits left after the decimal comma. So the speed of light in scientific notation, to two decimal places is $3,00 \times 10^8$ m $\cdot$ s^{-1}.
We can also perform addition, subtraction, multiplication and division with scientific notation. The following two worked examples show how to do this:

Example 1: *Addition and subtraction with scientific notation*

QUESTION

$1,99 \times 10^{-26} + 1,67 \times 10^{-27} - 2,79 \times 10^{-25} =?$

SOLUTION

Step 1 : ***Make all the exponents the same***

To add or subtract numbers in scientific notation we must make all the exponents the same:

$1,99 \times 10^{-26} = 0,199 \times 10^{-25}$ and

$1,67 \times 10^{-27} = 0,0167 \times 10^{-25}$

Step 2 : ***Carry out the addition and subtraction***

Now that the exponents are the same we can simply add or subtract the N part of each number:

$0,199 + 0,0167 - 2,79 = -2,5743$

Step 3 : ***Write the final answer***

To get the final answer we put the common exponent back:

$-2,5743 \times 10^{-25}$

Note that we follow the same process if the exponents are positive. For example $5,1 \times 10^3 + 4,2 \times 10^4 = 4,71 \times 10^4$.

Example 2: *Multiplication and division with scientific notation*

QUESTION

$1,6 \times 10^{-19} \times 3,2 \times 10^{-19} \div 5 \times 10^{-21}$

SOLUTION

Step 1 : ***Carry out the multiplication***

For multiplication and division the exponents do not need to be the same. For multiplication we add the exponents and multiply the N terms:

$1,6 \times 10^{-19} \times 3,2 \times 10^{-19} = (1,6 \times 3,2) \times 10^{-19+(-19)} = 5,12 \times 10^{-38}$

Step 2 : ***Carry out the division***

For division we subtract the exponents and divide the N terms. Using our result from the previous step we get:
$2,56 \times 10^{-38} \div 5 \times 10^{-21} = (5,12 \div 5) \times 10^{-38-(-19)} = 1,024 \times 10^{-18}$

Step 3 : ***Write the final answer***

The answer is: $1,024 \times 10^{-18}$

Note that we follow the same process if the exponents are positive. For example: $5,1 \times 10^3 \times 4,2 \times 10^4 = 21,42 \times 10^7 = 2,142 \times 10^8$

Units

ESAE

Imagine you had to make curtains and needed to buy fabric. The shop assistant would need to know how much fabric you needed. Telling her you need fabric 2 wide and 6 long would be insufficient — you have to specify the **unit** (i.e. 2 *metres* wide and 6 *metres* long). Without the unit the information is incomplete and the shop assistant would have to guess. If you were making curtains for a doll's house the dimensions might be 2 centimetres wide and 6 centimetres long!
It is not just lengths that have units, all physical quantities have units (e.g. time, temperature, distance, etc.).

DEFINITION: *Physical Quantity*

A physical quantity is anything that you can measure. For example, length, temperature, distance and time are physical quantities.

There are many different systems of units. The main systems of units are:

- SI units
- c.g.s units
- Imperial units
- Natural units

SI Units *ESAF*

We will be using the SI units in this course. SI units are the internationally agreed upon units.

DEFINITION: SI Units

The name *SI units* comes from the French *Système International d'Unités*, which means *international system of units*.

There are seven base SI units. These are listed in table 1.1. All physical quantities have units which can be built from these seven base units. So, it is possible to create a different set of units by defining a different set of base units.
These seven units are called base units because none of them can be expressed as combinations of the other six. These base units are like the 26 letters of the alphabet for English. Many different words can be formed by using these letters.

Base quantity	**Name**	**Symbol**
length	metre	m
mass	kilogram	kg
time	second	s
electric current	ampere	A
temperature	kelvin	K
amount of substance	mole	mol
luminous intensity	candela	cd

Table 1.1: SI Base Units

The Other Systems of Units *ESAG*

The SI Units are not the only units available, but they are most widely used. In Science there are three other sets of units that can also be used. These are mentioned here for interest only.

- **c.g.s. Units**
  In the c.g.s. system, the metre is replaced by the centimetre and the kilogram is replaced by the gram. This is a simple change but it means that all units derived from these two are changed. For example, the units of force and work are different. These units are used most often in astrophysics and atomic physics.

- **Imperial Units**
  Imperial units arose when kings and queens decided the measures that were to be used in the land. All the imperial base units, except for the measure of time, are different to those of SI units. This is the unit system you are most likely to encounter if SI units are not used. Examples of imperial units are pounds, miles, gallons and yards. These units are used by the Americans and British. As you can imagine, having different units in use from place to place makes scientific communication very difficult. This was the motivation for adopting a set of internationally agreed upon units.

- **Natural Units**
  This is the most sophisticated choice of units. Here the most fundamental discovered quantities (such as the speed of light) are set equal to 1. The argument for this choice is that all other quantities should be built from these fundamental units. This system of units is used in high energy physics and quantum mechanics.

Combinations of SI base units www ESAH

To make working with units easier, some combinations of the base units are given special names, but it is always correct to reduce everything to the base units. Table 1.2 lists some examples of combinations of SI base units that are assigned special names. Do not be concerned if the formulae look unfamiliar at this stage - we will deal with each in detail in the chapters ahead (as well as many others)!
It is very important that you are able to recognise the units correctly. For example, the **newton** (N) is another name for the **kilogram metre per second squared** ($kg{\cdot}m{\cdot}s^{-2}$), while the **kilogram metre squared per second squared** ($kg{\cdot}m^2{\cdot}s^{-2}$) is called the **joule** (J).

Quantity	Formula	Unit Expressed in Base Units	Name of Combination
Force	ma	kg · m · s^{-2}	N (newton)
Frequency	$\frac{1}{T}$	s^{-1}	Hz (hertz)
Work	Fs	kg · m^{2} · s^{-2}	J (joule)

Table 1.2: Some examples of combinations of SI base units assigned special names

Prefixes of base units

ESAI

Tip

When writing combinations of base SI units, place a dot (·) between the units to indicate that different base units are used. For example, the symbol for metres per second is correctly written as m·s^{-1}, and not as ms^{-1} or m/s. Although the last two options will be accepted in tests and exams, we will only use the first one in this book.

Now that you know how to write numbers in scientific notation, another important aspect of units is the prefixes that are used with the units. In the case of units, the prefixes have a special use. The kilogram (kg) is a simple example. 1 kg is equal to 1 000 g or 1×10^3 g. Grouping the 10^3 and the g together we can replace the 10^3 with the prefix k (kilo). Therefore the k takes the place of the 10^3. The kilogram is unique in that it is the only SI base unit containing a prefix.

In science, all the prefixes used with units are some power of 10. Table 1.3 lists some of these prefixes. You will not use most of these prefixes, but those prefixes listed in **bold** should be learnt. The case of the prefix symbol is very important. Where a letter features twice in the table, it is written in uppercase for exponents bigger than one and in lowercase for exponents less than one. For example M means mega (10^6) and m means milli (10^{-3}).

Prefix	**Symbol**	**Exponent**	**Prefix**	**Symbol**	**Exponent**
yotta	Y	10^{24}	yocto	y	10^{-24}
zetta	Z	10^{21}	zepto	z	10^{-21}
exa	E	10^{18}	atto	a	10^{-18}
peta	P	10^{15}	femto	f	10^{-15}
tera	T	10^{12}	pico	p	10^{-12}
giga	G	10^{9}	**nano**	n	10^{-9}
mega	M	10^{6}	**micro**	μ	10^{-6}
kilo	k	10^{3}	**milli**	m	10^{-3}
hecto	h	10^{2}	**centi**	c	10^{-2}
deca	da	10^{1}	**deci**	d	10^{-1}

Table 1.3: Unit Prefixes

Tip

There is no space and no dot between the prefix and the symbol for the unit.

Here are some examples of the use of prefixes:

- $40\ 000$ m can be written as 40 km (kilometre)
- $0,001$ g is the same as 1×10^{-3} g and can be written as 1 mg (milligram)
- $2,5 \times 10^{6}$ N can be written as $2,5$ MN (meganewton)
- $250\ 000$ A can be written as 250 kA (kiloampere) or $0,250$ MA (megaampere)
- $0,000000075$ s can be written as 75 ns (nanoseconds)
- 3×10^{-7} mol can be rewritten as $0,3 \times 10^{-6}$ mol, which is the same as $0,3$ μmol (micromol)

Exercise 1 - 1

1. Carry out the following calculations:
   a. $1,63 \times 10^{5} + 4,32 \times 10^{6} - 8,53 \times 10^{5}$
   b. $7,43 \times 10^{3} \div 6,54 \times 10^{7} \times 3,33 \times 10^{5}$
   c. $6,21434534 \times 10^{-5} \times 3,2555 \times 10^{-3} + 6,3 \times 10^{-4}$
2. Write the following in scientific notation using Table 1.3 as a reference.
   a. $0,511$ MV
   b. 10 cℓ
   c. $0,5$ μm

   d. 250 nm
   e. $0,00035$ hg

3. Write the following using the prefixes in Table 1.3.
   a. $1,602 \times 10^{-19}$ C
   b. $1,992 \times 10^{6}$ J
   c. $5,98 \times 10^{4}$ N
   d. 25×10^{-4} A
   e. $0,0075 \times 10^{6}$ m

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(1.) 02u4 (2.) 01v7 (3.) 01v8

The Importance of Units ESAJ

Without units much of our work as scientists would be meaningless. We need to express our thoughts clearly and units give meaning to the numbers we measure and calculate. Depending on which units we use, the numbers are different. For example if you have 12 water, it means nothing. You could have 12 ml of water, 12 litres of water, or even 12 bottles of water. Units are an essential part of the language we use. Units must be specified when expressing physical quantities. Imagine that you are baking a cake, but the units, like grams and millilitres, for the flour, milk, sugar and baking powder are not specified!

Group Discussion: Importance of Units

Work in groups of 5 to discuss other possible situations where using the incorrect set of units can be to your disadvantage or even dangerous. Look for examples at home, at school, at a hospital, when travelling and in a shop.

Case Study: The importance of units

Read the following extract from CNN News 30 September 1999 and answer the questions below.

NASA: Human error caused loss of Mars orbiter November 10, 1999

Failure to convert English measures to metric values caused the loss of the Mars Climate Orbiter, a spacecraft that smashed into the planet instead of reaching a safe orbit, a NASA investigation concluded Wednesday.

The Mars Climate Orbiter, a key craft in the space agency's exploration of the red planet, vanished on 23 September after a 10 month journey. It is believed that the craft came dangerously close to the atmosphere of Mars, where it presumably burned and broke into pieces.
An investigation board concluded that NASA engineers failed to convert English measures of rocket thrusts to newton, a metric system measuring rocket force. One English pound of force equals 4,45 newtons. A small difference between the two values caused the spacecraft to approach Mars at too low an altitude and the craft is thought to have smashed into the planet's atmosphere and was destroyed.
The spacecraft was to be a key part of the exploration of the planet. From its station about the red planet, the Mars Climate Orbiter was to relay signals from the Mars Polar Lander, which is scheduled to touch down on Mars next month.
"The root cause of the loss of the spacecraft was a failed translation of English units into metric units and a segment of ground-based, navigation-related mission software," said Arthur Stephenson, chairman of the investigation board.
Questions:

1. Why did the Mars Climate Orbiter crash? Answer in your own words.
2. How could this have been avoided?
3. Why was the Mars Orbiter sent to Mars?
4. Do you think space exploration is important? Explain your answer.

How to change units ESAK

It is very important that you are aware that different systems of units exist. Furthermore, you must be able to convert between units. Being able to change between units (for example, converting from millimetres to metres) is a useful skill in Science.
The following conversion diagrams will help you change from one unit to another.

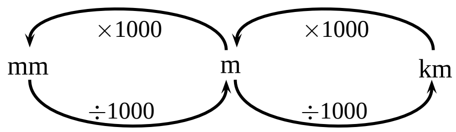


Figure 1.2: The distance conversion table

If you want to change millimetre to metre, you divide by 1000 (follow the arrow from mm to m); or if you want to change kilometre to millimetre, you multiply by 1000×1000.

The same method can be used to change millilitre to litre or kilolitre. Use Figure 1.3 to change volumes:

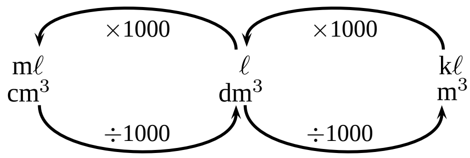


Figure 1.3: The volume conversion table

Example 3: *Conversion 1*

QUESTION

Express 3 800 mm in metres.

SOLUTION

Step 1 : ***Use the conversion table***

Use Figure 1.2. Millimetre is on the left and metre in the middle.

Step 2 : ***Decide which direction you are moving***

You need to go from mm to m, so you are moving from left to right.

Step 3 : ***Write the answer***

$3800 \text{ mm} \div 1\ 000 = 3,8 \text{ m}$

Example 4: *Conversion 2*

QUESTION

Convert 4,56 kg to g.

SOLUTION

Step 1 : ***Find the two units on the conversion diagram.***
Use Figure 1.2. Kilogram is the same as kilometre and gram is the same as metre.

Step 2 : ***Decide whether you are moving to the left or to the right.***
You need to go from kg to g, so it is from right to left.

Step 3 : ***Read from the diagram what you must do and find the answer.***
$4,56\ \text{kg} \times 1\ 000 = 4\ 560\ \text{g}$

Two other useful conversions

Very often in science you need to convert speed and temperature. The following two rules will help you do this:

1. **Converting speed**
When converting km $\cdot$ h^{-1} to m $\cdot$ s^{-1} you multiply by $1\ 000$ and divide by $3\ 600$ ($\frac{1000\ \text{m}}{3600\ \text{s}}$). For example 72 km $\cdot$ h^{-1} $\div 3,6 = 20$m $\cdot$ s^{-1}.
When converting m $\cdot$ s^{-1} to km $\cdot$ h^{-1}, you multiply by $3\ 600$ and divide by $1\ 000$ ($\frac{3600\ \text{s}}{1000\ \text{m}}$). For example 30 m $\cdot$ s^{-1} $\times 3,6 = 108$km $\cdot$ h^{-1}.
2. **Converting temperature**
Converting between the Kelvin and Celsius temperature scales is simple. To convert from Celsius to Kelvin add 273. To convert from Kelvin to Celsius subtract 273. Representing the Kelvin temperature by T_K and the Celsius temperature by $T_{^\circ C}$:

$$T_K = T_{^\circ C} + 273$$

Changing the subject of a formula ESAL

Very often in science you will have to change the subject of a formula. We will look at two examples. (Do not worry if you do not yet know what the terms and symbols mean, these formulae will be covered later in the book.)

1. **Moles**
   The equation to calculate moles from molar mass is: $n = \frac{m}{M}$, where n is the number of moles, m is the mass and M is the molar mass. As it is written we can easily find the number of moles of a substance. But what if we have the number of moles and want to find the molar mass? We note that we can simply multiply both sides of the equation by the molar mass and then divide both sides by the number of moles:

$$\begin{aligned} n &= \frac{m}{M} \\ nM &= m \\ M &= \frac{m}{n} \end{aligned}$$

   And if we wanted the mass we would use: $m = nM$.

2. Energy of a photon
   The equation for the energy of a photon is $E = h\frac{c}{\lambda}$, where E is the energy, h is Planck's constant, c is the speed of light and λ is the wavelength. To get c we can do the following:

$$\begin{aligned} E &= h\frac{c}{\lambda} \\ E\lambda &= hc \\ c &= \frac{E\lambda}{h} \end{aligned}$$

   Similarly we can find the wavelength we use: $\lambda = \frac{hc}{E}$ and to find Planck's constant we use: $h = \frac{E\lambda}{c}$.

Rate, proportion and ratios ESAM

In science we often want to know how a quantity relates to another quantity or how something changes over a period of time. To do this we need to know about rate, proportion and ratios.

Rate:

The rate at which something happens is the number of times that it happens over a period of time. The rate is always a change per time unit. So we can get rate of change of velocity per unit time ($\frac{\Delta \vec{v}}{t}$) or the rate of change in concentration per unit time (or $\frac{\Delta C}{t}$). (Note that Δ represents a change in).

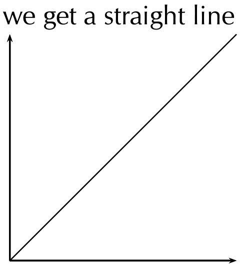
Ratios and fractions:

A fraction is a number which represents a part of a whole and is written as $\frac{a}{b}$, where a is the numerator and b is the denominator. A ratio tells us the relative size of one quantity (e.g. number of moles of reactants) compared to another quantity (e.g. number of moles of product): $2:1$, $4:3$, etc. Ratios can also be written as fractions as percentages (fractions with a denominator of 100).

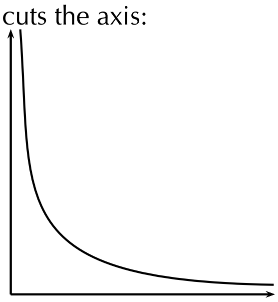
Proportion:

Proportion is a way of describing relationships between values or between constants. We can say that x is directly proportional to y ($x \propto y$) or that a is inversely proportional to b ($a \propto \frac{1}{b}$). It is important to understand the difference between directly and inversely proportional.

- **Directly proportional**
  Two values or constants are directly proportional when a change in one leads to the same change in the other. This is a more-more relationship. We can represent this as $y \propto x$ or $y = kx$ where k is the proportionality constant. We have to include k since we do not know by how much x changes when y changes. x could change by 2 for every change of 1 in y. If we plot two directly proportional variables on a graph, then we get a straight line graph that goes through the origin $(0;0)$:



- **Inversely proportional**
  Two values or constants are inversely proportional when a change in one leads to the opposite change in the other. We can represent this as $y = \frac{k}{x}$. This is a more-less relationship. If we plot two inversely proportional variables we get a curve that never cuts the axis:



Constants in equations

ESAN

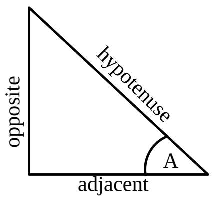
A constant in an equation **always** has the same value. For example the speed of light ($c = 2,99 \times 10^8$ m· s^{-1}), Planck's constant (h) and Avogadro's number (N_A) are all examples of constants that are use in science. The following table lists all the constants that you will encounter in this book.

Constant	Symbol	**Value and units**	**SI Units**
Atomic mass unit	u	$1,67 \times 10^{-24}$ g	$1,67 \times 10^{-27}$ kg
Charge on an electron	e	$-1,6 \times 10^{-19}$ C	$-1,6 \times 10^{-19}$ s · A
Speed of sound (in air, at 25°)		$344 m \cdot s^{-1}$	
Speed of light	c	$3 \times 10^8 m \cdot s^{-1}$	
Planck's constant	h	$6,626 \times 10^{-34}$ J · s	$6,626 \times 10^{-34}$ kg · m^{2}s^{-1}
Avogadro's number	N_A	$6,022 \times 10^{23}$	
Gravitational acceleration	g	$9,8$ m · s^{-1}	

Trigonometry

ESAO

Trigonometry is the relationship between the angles and sides of right angled triangles. Trigonometrical relationships are ratios and therefore have no units. You should know the following trigonometric ratios:



- Sine
  This is defined as $\sin A = \frac{\text{opposite}}{\text{hypotenuse}}$
- Cosine
  This is defined as $\cos A = \frac{\text{adjacent}}{\text{hypotenuse}}$

- Tangent
  This is defined as $\tan A = \frac{\text{opposite}}{\text{adjacent}}$

Exercise 1 - 2

1. Write the following quantities in scientific notation:
   a. 10130 Pa to 2 decimal places
   b. 978,15 m·s^{-2} to one decimal place
   c. 0,000001256 A to 3 decimal places
2. For each of the following symbols, write out the unit in full and write what power of 10 it represents:
   a. μg
   b. mg
   c. kg
   d. Mg
3. Write each of the following in scientific notation, correct to 2 decimal places:
   a. 0,00000123 N
   b. 417 000 000 kg
   c. 246800 A
   d. 0,00088 mm
4. For each of the following, write the measurement using the correct symbol for the prefix and the base unit:
   a. 1,01 microseconds
   b. 1 000 milligrams
   c. 7,2 megametres
   d. 11 nanolitre
5. The Concorde is a type of aeroplane that flies very fast. The top speed of the Concorde is 844 km·hr^{-1}. Convert the Concorde's top speed to m·s^{-1}.
6. The boiling point of water is 100 °C. What is the boiling point of water in kelvin?

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(1.) 01v9 (2.) 01va (3.) 01vb (4.) 01vc (5.) 01vd (6.) 01ve

Skills in the laboratory ESAP

To carry out experiments in the laboratory you need to know how to properly present your experimental results, you also need to know how to read instruments and how to interpret your data. A laboratory (be it for physics, chemistry or other sciences) can be a very dangerous and daunting place. However, if you follow a few simple guidelines you can safely carry out experiments in the laboratory without endangering yourself or others around you.

Experiments ESAQ

When a scientist performs experiments the following process is followed:

- Observe an event and identify an answerable question about the event.
- Make a hypothesis (theory) about the event that gives a sensible result.
- Design an experiment to test the theory. This includes identifying the fixed factors (what will not vary in the experiment), identifying the independent variable (this is set) and the dependent variable (what you will actually measure).
- Collect data accurately and interpret the data.
- Draw conclusions from the results of the experiment.
- Decide whether the hypothesis is correct or not.
- Verify your results by repeating the experiment or getting someone else to repeat the experiment.

This process is known as the scientific method. In the work that you will do you will be given the first three items and be required to determine the last four items. For verifying results you should see what your classmates obtained for their experiment.
In science the recording of practical work follows a specific layout. You should always present your work using this layout, as it will help any other person be able to understand and repeat your experiment.

- Aim: A brief sentence describing the purpose of the experiment.
- Apparatus: A sketch of the apparatus and a list of the apparatus

- Method: A list of the steps followed to carry out the experiment
- Results: Tables, graphs and observations about the experiment
- Discussion: What your results mean
- Conclusion: A brief sentence concluding whether or not the aim was met

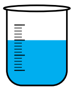
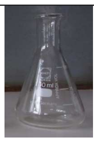
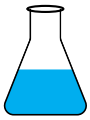
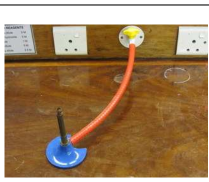
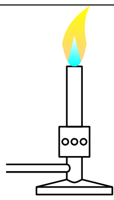
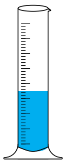
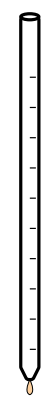
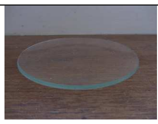
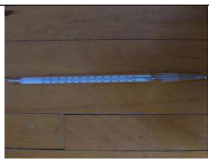
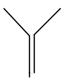
To perform experiments correctly and accurately you also need to know how to work with various pieces of equipment. The next section details some of the apparatus that you need to know, as well as how to correctly work with it.

As you work through the experiments in the book you will be given guidance on how to present your data. By the end of the year you should be able to select the appropriate method to show your data, whether it is a table, a graph or an equation.

You will also need to know how to interpret your data. For example given a table of values, what can you say about those values. Also you should be able to say whether you are performing a qualitative (descriptive) or a quantitative (numbers) analysis.

Laboratory apparatus ESAR

Listed here are some of the common pieces of apparatus that you will be working with in the laboratory. You should be able to name all the apparatus listed here as well as make a simple sketch of it.

Item	**Photo**	**Sketch**
Beaker		
Flask		
Test tubes		
Bunsen burner		
Measuring cylinder		
Pipette		
Watch glass		
Thermometer		
Funnel		

The following image shows the correct setup for heating liquids on a Bunsen burner:

When reading any instrument (such as a measuring cylinder, a pipette, etc.) always make sure that the instrument is level and that your eye is at the level of the top of the liquid.

General safety rules

ESAS

The following are some of the general guidelines and rules that you should always observe when working in a laboratory.

1. You are responsible for your own safety as well as the safety of others in the laboratory.
2. Do not eat or drink in the laboratory. Do not use laboratory glassware to eat or drink from.
3. Always behave responsibly in the laboratory. Do not run around or play practical jokes.
4. In case of accidents or chemical spills call your teacher at once.
5. Always check with your teacher how to dispose of waste. Chemicals should not be disposed of down the sink.
6. Only perform the experiments that your teacher instructs you to. Never mix chemicals for fun.
7. Never perform experiments alone.
8. Always check the safety data of any chemicals you are going to use.
9. Follow the given instructions exactly. Do not mix up steps or try things in a different order.
10. Be alert and careful when handling chemicals, hot glassware, etc.
11. Ensure all Bunsen burners are turned off at the end of the practical and all chemical containers are sealed.
12. Never add water to acid. Always add the acid to water.
13. Never heat thick glassware as it will break. (i.e. do not heat measuring cylinders).
14. When you are smelling chemicals, place the container on a laboratory bench and

use your hand to gently waft (fan) the vapours towards you.

15. Do not take chemicals from the laboratory.

16. Always work in a well ventilated room. Whenever you perform experiments, you should open the windows.

17. Do not leave Bunsen burners and flames unattended.

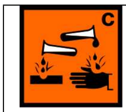
18. Never smell, taste or touch chemicals unless instructed to do so.

19. Never point test tubes at people or yourself. When heating chemicals, always point the mouth of the test tube away from you and your classmates.

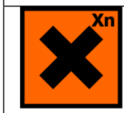
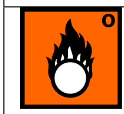
Hazard signs

ESAT

The table below lists some of the common hazards signs that you may encounter. You should know what all of these mean.

Sign	Symbol	Meaning	Sign	Symbol	Meaning
	C	Corrosive. Chemicals with this label can burn your skin and eyes and burn holes in your clothes. An example is hydrochloric acid.		N	Environmentally harmful. Chemicals with this label are damaging to the environment. An example is CFC's.

continued on next page

Sign	Symbol	Meaning	Sign	Symbol	Meaning
	E	Explosive. Chemicals with this label explode easily. An example is lead azide.		F	Flammable. Chemicals with this label can catch fire easily. Example: methanol
	Xn	Harmful. Chemicals labelled with this are generally considered to be damaging to humans.		Xi	Irritant. Chemicals with this label cause irritation to your eyes and skin. An example is hydrogen peroxide.
	O	Oxidising. Chemicals with this label contain oxygen that may cause other materials to combust. An example is potassium dichromate.		T	Toxic. Chemicals with this label are highly toxic. An example is mercury.

Notes and information ESAU

You can find safety data sheets at *http://www.msds.com/*. You should always look at these data sheets anytime you work with a new chemical. These data sheets contain information about how to work with chemicals and what dangers the chemicals pose to you and the environment. You should always try dispose of chemicals correctly and safely. Many chemicals cannot simply be washed down the sink.

Classification of matter 2

Materials

ESAV

All the objects that we see in the world around us, are made of **matter**. Matter makes up the air we breathe, the ground we walk on, the food we eat and the animals and plants that live around us. Even our own human bodies are made of matter!

Different objects can be made of different types of **materials** (the matter from which objects are made). For example, a cupboard (an *object*) is made of wood, nails, hinges and knobs (the *materials*). The **properties** of the materials will affect the properties of the object. In the example of the cupboard, the strength of the wood and metals make the cupboard strong and durable. It is very important to understand the properties of materials, so that we can use them in our homes, in industry and in other applications.

Cupboard

Photo by grongar on Flickr.com

See introductory video: (Video: VPabo at www.everythingscience.co.za)

Some of the properties of matter that you should know are:

- Materials can be **strong** and resist bending (e.g. bricks, rocks) or **weak** and bend easily (e.g. clothes)
- Materials that conduct heat (e.g. metals) are called **thermal conductors**. Materials that conduct electricity (e.g. copper wire) are **electrical conductors**.
- **Brittle** materials break easily (e.g. plastic). Materials that are **malleable** can be easily formed into different shapes (e.g. clay, dough). **Ductile** materials are able to be formed into long wires (e.g. copper).
- **Magnetic** materials have a magnetic field (e.g. iron).
- **Density** is the mass per unit volume. Examples of dense materials include concrete and stones.
- The **boiling and melting points** of substances tells us the temperature at which the substance will boil or melt. This helps us to classify substances as solids, liquids or gases at a specific temperature.

The diagram below shows one way in which matter can be classified (grouped) according to its different properties. As you read further in this chapter, you will see that there are also other ways of classifying materials, for example according to whether or not they are good electrical conductors.

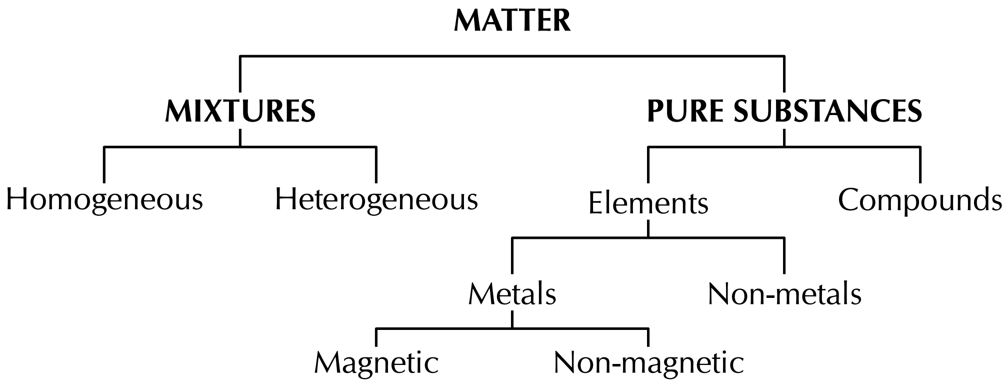


Figure 2.2: The classification of matter

Activity:	What materials are products made of?

This activity looks at the materials that make up food products. In groups of 3 or 4 look at the labels on food items. Make a list of the ingredients. Can you tell from the ingredients what the food is (i.e. spice, oil, sweets, etc.)? Food products are labelled to help you (the consumer) know what you are eating and to help you choose healthier alternatives. Some compounds, such as MSG and tartrazine are being removed from products due to being regarded as unsafe. Are there other ingredients in the products that are unsafe to eat? What preservatives and additives (e.g. tartrazine, MSG, colourants) are there? Are these preservatives and additives good for you? Are there natural (from plants) alternatives? What do different indigenous people groups use to flavor and preserve their food?

Some labels on food items

Activity:	*Classifying materials*

Look around you at the various structures. Make a list of all the different materials that you see. Try to work out why a particular material was used. Can you classify all the different materials used according to their properties? Why are these materials chosen over other materials?

Picture by flowcomm on Flickr.com

Mixtures

ESAW

We see mixtures all the time in our everyday lives. A stew, for example, is a mixture of different foods such as meat and vegetables; sea water is a mixture of water, salt and other substances, and air is a mixture of gases such as carbon dioxide, oxygen and nitrogen.

> *DEFINITION: Mixture*
>
> A mixture is a combination of two or more substances, where these substances are not bonded (or joined) to each other and no chemical reaction occurs between the substances.

In a mixture, the substances that make up the mixture:

- **are not in a fixed ratio**
  Imagine, for example, that you have 250 ml of water and you add sand to the water. It doesn't matter whether you add 20 g, 40 g, 100 g or any other mass of sand to the water; it will still be called a mixture of sand and water.
- **keep their physical properties**
  In the example we used of sand and water, neither of these substances has changed in any way when they are mixed together. The sand is still sand and the water is still water.
- **can be separated by mechanical means**
  To separate something by "mechanical means", means that there is no chemical process involved. In our sand and water example, it is possible to separate the mixture by

simply pouring the water through a filter. Something *physical* is done to the mixture, rather than something *chemical*.

We can group mixtures further by dividing them into those that are heterogeneous and those that are homogeneous.

Heterogeneous mixtures ESAX

A **heterogeneous** mixture does not have a definite composition. Cereal in milk is an example of a heterogeneous mixture. Soil is another example. Soil has pebbles, plant matter and sand in it. Although you may add one substance to the other, they will stay separate in the mixture. We say that these heterogeneous mixtures are *non-uniform*, in other words they are not exactly the same throughout.

Cereal

Picture by dougww on Flickr.com

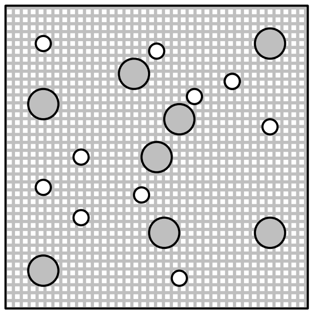


Figure 2.3: A submicroscopic representation of a heterogeneous mixture. The gray circles are one substance (e.g. one cereal) and the white circles are another substance (e.g. another cereal). The background is the milk.

> ### *DEFINITION: Heterogeneous mixture*
>
> A heterogeneous mixture is one that consists of two or more substances. It is non-uniform and the different components of the mixture can be seen.

Heterogeneous mixtures can be further subdivided according to whether it is two liquids mixed, a solid and a liquid or a liquid and a gas or even a gas and a solid. These mixtures are given special names which you can see in table below.

Phases of matter	Name of mixture	Example
liquid-liquid	emulsion	oil in water
solid-liquid	suspension	muddy water
gas-liquid	aerosol	fizzy drinks
gas-solid	smoke	smog

Table 2.1: Examples of different heterogeneous mixtures

Homogeneous mixtures

ESAY

A **homogeneous** mixture has a definite composition, and specific properties. In a homogeneous mixture, the different parts cannot be seen. A solution of salt dissolved in water is an example of a homogeneous mixture. When the salt dissolves, it spreads evenly through the water so that all parts of the solution are the same, and you can no longer see the salt as being separate from the water. Think also of coffee without milk. The air we breathe is another example of a homogeneous mixture since it is made up of different gases which are in a constant ratio, and which can't be visually distinguished from each other (i.e. you can't see the different components).

See video: VPabz at www.everythingscience.co.za

Coffee

Photo by Julius Schorzman on Wikimedia

Salt dissolving in water

DEFINITION: Homogeneous mixture

A homogeneous mixture is one that is uniform, and where the different components of the mixture cannot be seen.

FACT

An **alloy** is a homogeneous mixture of two or more elements, at least one of which is a metal, where the resulting material has metallic properties. For example steel is an alloy made up mainly from iron with a small amount of carbon (to make it harder), manganese (to make it strong) and chromium (to prevent rusting).

Example 1: Mixtures

QUESTION

For each of the following mixtures state whether it is a homogeneous or a heterogeneous mixture:

a. *sugar dissolved in water*
b. *flour and iron filings (small pieces of iron)*

SOLUTION

Step 1 : ***Look at the definition***

We first look at the definition of a heterogeneous and homogeneous mixture.

Step 2 : ***Decide whether or not you can see the components***

a. We cannot see the sugar in the water.
b. We are able to make out the pieces of iron in the flour.

Step 3 : ***Decide whether or not the components are mixed uniformly***

a. The two components are mixed uniformly.
b. In this mixture there may be places where there are a lot of iron filings and places where there is more flour, so it is not uniformly mixed.

Step 4 : ***Give the final answer***

a. Homogeneous mixture.
b. Heterogeneous mixture.

Activity: *Making mixtures*

Make mixtures of sand and water, potassium dichromate and water, iodine and ethanol, iodine and water. Classify these as heterogeneous or homogeneous. Give reasons for your choice.

Make your own mixtures by choosing any two substances from

- sand
- water
- stones
- cereal
- salt
- sugar

Try to make as many different mixtures as possible. Classify each mixture and give a reason for your choice.

Potassium dichromate (top) and iodine (bottom)

Exercise 2 - 1

Complete the following table:

Substance	Non-mixture or mixture	Heterogeneous mixture	Homogeneous mixture
tap water			
brass (an alloy of copper and zinc)			
concrete			
aluminium foil (tinfoil)			
Coca Cola			
soapy water			
black tea			
sugar water			
baby milk formula			

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Pure substances

ESAZ

Any material that is not a mixture, is called a **pure substance**. Pure substances include **elements** and **compounds**. It is much more difficult to break down pure substances into their parts, and complex chemical methods are needed to do this.

See video: VPacc at www.everythingscience.co.za

We can use melting and boiling points and chromatography to test for pure substances. Pure substances have a sharply defined (one temperature) melting or boiling point. Impure substances have a temperature range over which they melt or boil. Chromatography is the process of separating substances into their individual components. If a substance is pure then chromatography will only produce one substance at the end of the process. If a substance is impure then several substances will be seen at the end of the process.

Activity: *Recommended practical activity: Smartie chromatography*

You will need:

- filter paper (or blotting paper)
- some smarties in different colours
- water
- an eye dropper.

Place a smartie in the centre of a piece of filter paper. Carefully drop a few drops of water onto the smartie, until the smartie is quite wet and there is a ring of water on the filter paper. After some time you should see a coloured ring on the paper around the smartie. This is because the food colouring that is used to make the smartie colourful dissolves in the water and is carried through the paper away from the smartie.

Smartie chromatography

Photo by Neil Ravenscroft - UCT

Elements

ESAAA

An **element** is a chemical substance that can't be divided or changed into other chemical substances by any ordinary chemical means. The smallest unit of an element is the **atom**.

> *DEFINITION: Element*
>
> An element is a substance that cannot be broken down into other substances through chemical means.

FACT

Recently it was agreed that two more elements would be added to the list of officially named elements. These are elements number 114 and 116. The proposed name for element 114 is flerovium and for element 116 it is moscovium. This brings the total number of officially named elements to 114.

There are 112 officially named elements and about 118 known elements. Most of these are natural, but some are man-made. The elements we know are represented in the **periodic table**, where each element is abbreviated to a **chemical symbol**. Table 2.3 gives the first 20 elements and some of the common transition metals.

Element name	**Element symbol**	**Element name**	**Element symbol**
Hydrogen	H	Phosphorus	P
Helium	He	Sulphur	S
Lithium	Li	Chlorine	Cl
Beryllium	Be	Argon	Ar
Boron	B	Potassium	K
Carbon	C	Calcium	Ca
Nitrogen	N	Iron	Fe
Oxygen	O	Nickel	Ni
Fluorine	F	Copper	Cu
Neon	Ne	Zinc	Zn
Sodium	Na	Silver	Ag
Magnesium	Mg	Platinum	Pt
Aluminium	Al	Gold	Au
Silicon	Si	Mercury	Hg

Table 2.2: List of the first 20 elements and some common transition metals

Compounds ESAAB

A **compound** is a chemical substance that forms when two or more different elements combine in a fixed ratio. Water (H_2O), for example, is a compound that is made up of two hydrogen atoms for every one oxygen atom. Sodium chloride (NaCl) is a compound made up of one sodium atom for every chlorine atom. An important characteristic of a compound is that it has a **chemical formula**, which describes the ratio in which the atoms of each element in the compound occur.

DEFINITION: Compound

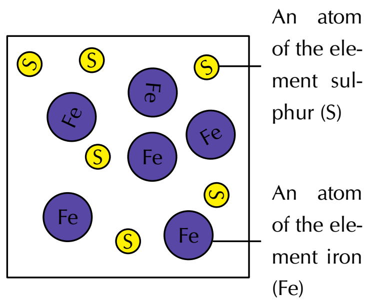
A substance made up of two or more different elements that are joined together in a fixed ratio.

See video: VPacw at www.everythingscience.co.za

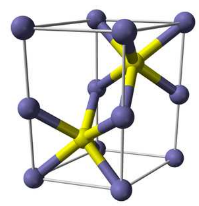
Figure 2.3 might help you to understand the difference between the terms **element**, **mixture** and **compound**. Iron (Fe) and sulphur (S) are two elements. When they are added together, they form a *mixture* of iron and sulphur. The iron and sulphur are not joined together. However, if the mixture is heated, a new **compound** is formed, which is called iron sulphide (FeS).

Note

Figure 2.3 showed a submicroscopic representation of a mixture. In a submicroscopic representation we use circles to represent different elements. To show a compound, we draw several circles joined together. Mixtures are simply shown as two or more individual elements in the same box. The circles are not joined for a mixture.



A mixture of iron and sulphur



A model of the iron sulphide crystal

We can also use symbols to represent elements, mixtures and compounds. The symbols for the elements are all found on the periodic table. Compounds are shown as two or more element names written right next to each other. Subscripts may be used to show that there is more than one atom of a particular element. (e.g. H_2O or NH_3). Mixtures are written as: a mixture of element (or compound) A and element (or compound) B. (e.g. a mixture of Fe and S).

Example 2: *Mixtures and pure substances*

QUESTION

For each of the following substances state whether it is a pure substance or a mixture. If it is a mixture, is it homogeneous or heterogeneous? If it is a pure substance is it an element or a compound?

a. *Blood (which is made up from plasma and cells)*
b. *Argon*
c. *Silicon dioxide (SiO_2)*
d. *Sand and stones*

SOLUTION

Step 1 : ***Apply the definitions***

An element is found on the periodic table, so we look at the periodic table and find that only argon appears there. Next we decide which are compounds and which are mixtures. Compounds consist of two or more elements joined in a fixed ratio. Sand and stones are not elements, neither is blood. But silicon is, as is oxygen. Finally we decide whether the mixtures are homogeneous or heterogeneous. Since we cannot see the separate components of blood it is homogeneous. Sand and stones are heterogeneous.

Step 2 : ***Write the answer***

a. Blood is a homogeneous mixture.
b. Argon is a pure substance. Argon is an element.
c. Silicon dioxide is a pure substance. It is a compound.
d. Sand and stones form a heterogeneous mixture.

Activity: *Using models to represent substances*

The following substances are given:

- Air (consists of oxygen, nitrogen, hydrogen, water vapour)
- Hydrogen gas (H_2)
- Neon gas

- Steam
- Ammonia gas (NH_3)

1. Use coloured balls to build models for each of the substances given.
2. Classify the substances according to elements, compounds, homogeneous mixtures, heterogeneous mixture, pure substance, impure substance.
3. Draw submicroscopic representations for each of the above examples.

Exercise 2 - 2

1. In the following table, tick whether each of the substances listed is a *mixture* or a *pure substance*. If it is a mixture, also say whether it is a homogeneous or heterogeneous mixture.

Substance	Mixture or pure	Homogeneous or heterogeneous mixture
fizzy colddrink		
steel		
oxygen		
iron filings		
smoke		
limestone ($CaCO_3$)		

2. In each of the following cases, say whether the substance is an element, a mixture or a compound.
   a. Cu
   b. iron and sulphur
   c. Al
   d. H_2SO_4
   e. SO_3

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(1.) 0001 (2.) 0002

Names and formulae of substances

ESAAC

Think about what you call your friends. Some of your friends might have full names (long names) and a nickname (short name). These are the words we use to tell others who or what we are referring to. Their full name is like the substances name and their nickname is like the substances formulae. Without these names your friends would have no idea which of them you are referring to. Chemical substances have names, just like people have names. This helps scientists to communicate efficiently.

It is easy to describe elements and mixtures. We simply use the names that we find on the periodic table for elements and we use words to describe mixtures. But how are compounds named? In the example of iron sulphide that was used earlier, the compound name is a combination of the names of the elements but slightly changed.

See video: VPadm at www.everythingscience.co.za

The following are some guidelines for naming compounds:

1. The compound name will always include the **names of the elements** that are part of it.
   - A compound of **iron** (Fe) and *sulphur* (S) is **iron** *sulph*ide (FeS)
   - A compound of **potassium** (K) and *bromine* (Br) is **potassium** *brom*ide (KBr)
   - A compound of **sodium** (Na) and *chlorine* (Cl) is **sodium** *chlor*ide (NaCl)
2. In a compound, the element that is on the left of the Periodic Table, is used *first* when naming the compound. In the example of NaCl, sodium is a group 1 element on the left hand side of the table, while chlorine is in group 7 on the right of the table. Sodium therefore comes first in the compound name. The same is true for FeS and KBr.
3. The **symbols** of the elements can be used to represent compounds e.g. FeS, NaCl, KBr and H_2O. These are called **chemical formulae**. In the first three examples, the ratio of the elements in each compound is 1:1. So, for FeS, there is one atom of iron for every atom of sulphur in the compound. In the last example (H_2O) there are two atoms of hydrogen for every atom of oxygen in the compound.
4. A compound may contain **ions** (an ion is an atom that has lost or gained electrons). These ions can either be simple (consist of only one element) or compound (consist of several elements). Some of the more common ions and their formulae are given in Table 2.3 and in Table 2.4. You should know all these ions.

Compound ion	Formula	Compound ion	Formula	Compound ion	Formula
Hydrogen	H^+	Lithium	Li^+	Sodium	Na^+
Potassium	K^+	Silver	Ag^+	Mercury (I)	Hg^+
Copper (I)	Cu^+	Ammonium	NH_4^+	Beryllium	Be^{2+}
Magnesium	Mg^{2+}	Calcium	Ca^{2+}	Barium	Ba^{2+}
Tin (II)	Sn^{2+}	Lead (II)	Pb^{2+}	Chromium (II)	Cr^{2+}
Manganese (II)	Mn^{2+}	Iron (II)	Fe^{2+}	Cobalt (II)	Co^{2+}
Nickel	Ni^{2+}	Copper (II)	Cu^{2+}	Zinc	Zn^{2+}
Aluminium	Al^{3+}	Chromium (III)	Cr^{3+}	Iron (III)	Fe^{3+}
Cobalt (III)	Co^{3+}	Chromium (VI)	Cr^{6+}	Manganese (VII)	Mn^{7+}

Table 2.3: Table of cations

Compound ion	Formula	Compound ion	Formula
Fluoride	F^-	Oxide	O^{2-}
Chloride	Cl^-	Peroxide	O_2^{2-}
Bromide	Br^-	Carbonate	CO_3^{2-}
Iodide	I^-	Sulphide	S^{2-}
Hydroxide	OH^-	Sulphite	SO_3^{2-}
Nitrite	NO_2^-	Sulphate	SO_4^{2-}
Nitrate	NO_3^-	Thiosulphate	$S_2O_3^{2-}$
Hydrogen carbonate	HCO_3^-	Chromate	CrO_4^{2-}
Hydrogen sulphite	HSO_3^-	Dichromate	$Cr_2O_7^{2-}$
Hydrogen sulphate	HSO_4^-	Manganate	MnO_4^{2-}
Dihydrogen phosphate	$H_2PO_4^-$	Oxalate	$(COO)_2^{2-}/C_2O_4^{2-}$
Hypochlorite	ClO^-	Hydrogen phosphate	HPO_4^{2-}
Chlorate	ClO_3^-	Nitride	N^{3-}
Permanganate	MnO_4^-	Phosphate	PO_4^{3-}
Acetate (ethanoate)	CH_3COO^-	Phosphide	P^{3-}

Table 2.4: Table of anions

5. **Prefixes** can be used to describe the ratio of the elements that are in the compound.

This is used for non-metals. For metals, we add a roman number (I, II, III, IV) in brackets after the metal ion to indicate the ratio. You should know the following prefixes: “mono” (one), “di” (two) and “tri” (three).

- CO (carbon **mon**oxide) - There is one atom of oxygen for every one atom of carbon
- NO_2 (nitrogen **di**oxide) - There are two atoms of oxygen for every one atom of nitrogen
- SO_3 (sulphur **tri**oxide) - There are three atoms of oxygen for every one atom of sulphur

Tip

When numbers are written as “subscripts” in compounds (i.e. they are written below and to the right of the element symbol), this tells us how many atoms of that element there are in relation to other elements in the compound. For example in nitrogen dioxide (NO_2) there are two oxygen atoms for every one atom of nitrogen. Later, when we start looking at chemical equations, you will notice that sometimes there are numbers *before* the compound name. For example, $2H_2O$ means that there are two molecules of water, and that in each molecule there are two hydrogen atoms for every one oxygen atom.

The above guidelines also help us to work out the formula of a compound from the name of the compound. The following worked examples will look at names and formulae in detail.

We can use these rules to help us name both ionic compounds and covalent compounds. However, covalent compounds are often given other names by scientists to simplify the name (or because the molecule was named long before its formula was discovered). For example, if we have 2 hydrogen atoms and one oxygen atom the above naming rules would tell us that the substance is dihydrogen monoxide. But this compound is better known as water!
Some common covalent compounds are given in table 2.4

Name	Formula	Name	Formula
water	H_2O	hydrochloric acid	HCl
sulphuric acid	H_2SO_4	methane	CH_4
ethane	C_2H_6	ammonia	NH_3
nitric acid	HNO_3		

Table 2.5: Names of common covalent compounds

Example 3: *Writing chemical formulae 1*

QUESTION

What is formula of sodium fluoride?

SOLUTION

Step 1 : ***List the ions involved:***
We have the sodium ion (Na^+) and the fluoride ion (F^-). (You can look these up on the tables of cations and anions.)

Step 2 : ***Write down the charges on the ions***

The sodium ion has a charge of $+1$ and the fluoride ion has a charge of -1.

Step 3 : ***Find the right combination***

For every plus, we must have a minus. So the $+1$ from sodium cancels out the -1 from fluoride. They combine in a $1:1$ ratio.

Step 4 : ***Write the formula***

NaF

Example 4: *Writing chemical formulae 2*

QUESTION

What is the formula for magnesium chloride?

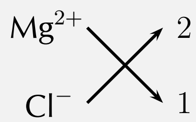
SOLUTION

Step 1 : ***List the ions involved***

Mg^{2+} and Cl^{-}

Step 2 : ***Find the right combination***

Magnesium has a charge of $+2$ and would need two chlorides to balance the charge. They will combine in a 1:2 ratio. There is an easy way to find this ratio:



Draw a cross as above, and then you can see that Mg $\rightarrow$ 1 and Cl $\rightarrow$ 2.

Step 3 : ***Write down the formula***

$MgCl_2$

Example 5: *Writing chemical formulae 3*

QUESTION

Write the chemical formula for magnesium oxide.

SOLUTION

Step 1 : ***List the ions involved.***

Mg^{2+} and O^{2-}

Step 2 : ***Find the right combination***

Mg^{2+} : 2

O^{2-} : 2

If you use the cross method, you will get a ratio of 2 : 2. This ratio must always be in simplest form, i.e. 1 : 1.

Step 3 : ***Write down the formula***

MgO (**not** Mg_2O_2)

Example 6: *Writing chemical formulae 4*

QUESTION

Write the formula for copper(II) nitrate.

SOLUTION

Step 1 : ***List the ions involved***

Cu^{2+} (the questions asks for copper(II) not copper(I))

NO_3^-

Step 2 : ***Find the right combination***

$$\begin{array}{ccc} Cu^{2+} & \times & 2 \\ NO_3^{-} & & 1 \end{array}$$

Step 3 : ***Write the formula***

$Cu(NO_3)_2$

Tip

Notice how in the last example we wrote NO_3^- inside brackets. We do this to indicate that NO_3^- is a compound ion and that there are two of these ions bonded to one copper ion.

Activity: *The ions dating game*

Your teacher will assign each of you a different ion (written on a piece of card). Stick this to yourself. You will also get cards with the numbers 1 - 5 on them. Now walk around the class and try to work out who you can pair up with and in what ratio. Once you have found a partner, indicate your ratio using the numbered cards. Check your results with your classmates or your teacher.

Exercise 2 - 3

1. The formula for calcium carbonate is $CaCO_3$.
   - **a**. Is calcium carbonate an element or a compound? Give a reason for your answer.
   - **b**. What is the ratio of Ca : C : O atoms in the formula?
2. Give the name of each of the following substances.
   - **a**. KBr
   - **b**. HCl
   - **c**. $KMnO_4$
   - **d**. NO_2
   - **e**. NH_4OH
   - **f**. Na_2SO_4
   - **g**. $Fe(NO_3)_3$
   - **h**. $PbSO_3$
   - **i**. $Cu(HCO_3)_2$
3. Give the chemical formula for each of the following compounds.

a. potassium nitrate
b. sodium oxide
c. barium sulphate
d. aluminium chloride
e. magnesium phosphate
f. tin(II) bromide
g. manganese(II) phosphide

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(1.) 0003 (2.) 0004 (3.) 0005

Metals, Metalloids and Non-metals

ESAAD

The elements in the periodic table can also be divided according to whether they are **metals**, **metalloids** or **non-metals**. The zigzag line separates all the elements that are metals from those that are non-metals. Metals are found on the left of the line, and non-metals are those on the right. Along the line you find the metalloids. You should notice that there are more metals then non-metals. Metals, metalloids and non-metals all have their own specific properties.

See video: VPaec at www.everythingscience.co.za

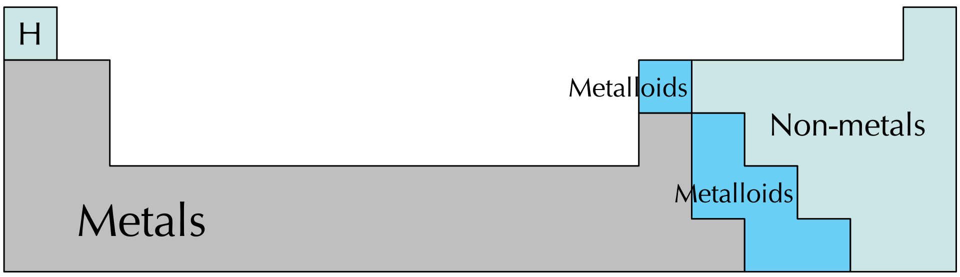


Figure 2.4: A simplified diagram showing part of the periodic table.

Metals

ESAAE

Examples of metals include copper (Cu), zinc (Zn), gold (Au), silver (Ag), tin (Sn) and lead (Pb). The following are some of the properties of metals:

Copper wire

- *Thermal conductors*
  Metals are good conductors of heat and are therefore used in cooking utensils such as pots and pans.
- *Electrical conductors*
  Metals are good conductors of electricity, and are therefore used in electrical conducting wires.
- *Shiny metallic lustre*
  Metals have a characteristic shiny appearance and are often used to make jewellery.
- *Malleable and ductile*
  This means that they can be bent into shape without breaking (malleable) and can be stretched into thin wires (ductile) such as copper.
- *Melting point*
  Metals usually have a high melting point and can therefore be used to make cooking pots and other equipment that needs to become very hot, without being damaged.
- *Density*
  Metals have a high density.
- *Magnetic properties*
  Only three main metals (iron, cobalt and nickel) are magnetic, the others are non-magnetic.

You can see how the properties of metals make them very useful in certain applications.

Activity:	*Group Work : Looking at metals*

1. Collect a number of metal items from your home or school. Some examples are listed below:
   - hair clips
   - safety pins
   - cooking pots
   - jewellery
   - scissors
   - cutlery (knives, forks, spoons)
2. In groups of 3-4, combine your collection of metal objects.
3. What is the function of each of these objects?
4. Discuss why you think metal was used to make each object. You should consider the properties of metals when you answer this question.

Non-metals ESAAF

In contrast to metals, non-metals are poor thermal conductors, good electrical insulators (meaning that they do *not* conduct electrical charge) and are neither malleable nor ductile. The non-metals include elements such as sulphur (S), phosphorus (P), nitrogen (N) and oxygen (O).

Sulphur

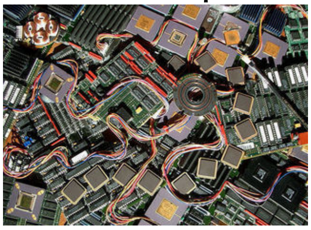
Picture by nickstone333 on Flickr.com

Metalloids ESAAG

Metalloids or semi-metals have mostly non-metallic properties. One of their distinguishing characteristics is that their conductivity increases as their temperature increases.

This is the opposite of what happens in metals. This property is known as semi-conductance and the materials are called semi-conductors. Semi-conductors are important in digital electronics, such as computers. The metalloids include elements such as silicon (Si) and germanium (Ge).

Silicon chips



Picture by jurveston on Flickr.com

Electrical conductors, semi-conductors and insulators

ESAAH

> **DEFINITION: Electrical conductor**
>
> An electrical conductor is a substance that allows an electrical current to pass through it.

Electrical conductors are usually metals. *Copper* is one of the best electrical conductors, and this is why it is used to make conducting wire. In reality, *silver* actually has an even higher electrical conductivity than copper, but silver is too expensive to use.

See video: VPaex at www.everythingscience.co.za

In the overhead power lines that we see above us, *aluminium* is used. The aluminium usually surrounds a steel core which adds makes it stronger so that it doesn't break when it is stretched across distances. Sometimes gold is used to make wire because it is very resistant to surface corrosion. *Corrosion* is when a material starts to deteriorate because of its reactions with oxygen and water in the air.

Power lines

Picture by Tripp on Flickr.com

> **DEFINITION: Insulators**
>
> An insulator is a non-conducting material that does not carry any charge.

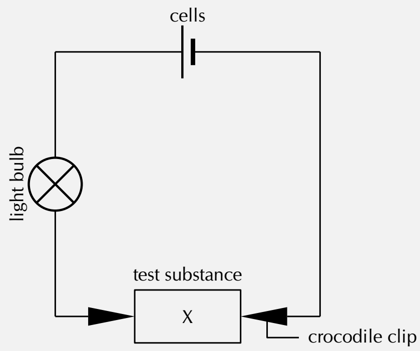
Examples of insulators are plastic and wood. **Semi-conductors** behave like insulators when they are cold, and like conductors when they are hot. The elements silicon and germanium are examples of semi-conductors.

General experiment: *Electrical conductivity*

Aim:
To investigate the electrical conductivity of a number of substances

Apparatus:

- two or three cells
- light bulb
- crocodile clips
- wire leads
- a selection of test substances (e.g. a piece of plastic, aluminium can, metal pencil sharpener, magnet, wood, chalk, cloth).



Method:

1. Set up the circuit as shown above, so that the test substance is held between the two crocodile clips. The wire leads should be connected to the cells and the light bulb should also be connected into the circuit.
2. Place the test substances one by one between the crocodile clips and see what happens to the light bulb. If the light bulb shines it means that current is flowing and the substance you are testing is an **electrical conductor**.

Results:
Record your results in the table below:

Test substance	Metal/non-metal	Does the light bulb glow?	Conductor or insulator

Conclusions:
In the substances that were tested, the metals were able to conduct electricity and the non-metals were not. Metals are good electrical conductors and non-metals are not.

See simulation: (Simulation: VPcyz at www.everythingscience.co.za)

Thermal Conductors and Insulators

ESAAI

A **thermal conductor** is a material that allows energy in the form of heat, to be transferred within the material, without any movement of the material itself. An easy way to understand this concept is through a simple demonstration.

See video: VPafb at www.everythingscience.co.za

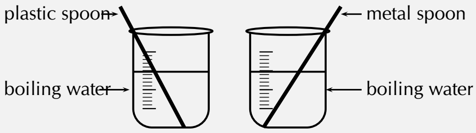
General experiment: *Demonstration: Thermal conductivity*

Aim:
To demonstrate the ability of different substances to conduct heat.

Apparatus:

You will need:

- two cups (made from the same material e.g. plastic)
- a metal spoon
- a plastic spoon.



Method:

- Pour boiling water into the two cups so that they are about half full.
- Place a metal spoon into one cup and a plastic spoon in the other.
- Note which spoon heats up more quickly

Warning:

Be careful when working with boiling water and when you touch the spoons as you can easily burn yourself.

Results:
The metal spoon heats up faster than the plastic spoon. In other words, the metal conducts heat well, but the plastic does not.

Conclusion: Metal is a good thermal conductor, while plastic is a poor thermal conductor.

FACT

Well-insulated buildings need less energy for heating than buildings that have no insulation. Two building materials that are being used more and more worldwide, are **mineral wool** and **polystyrene**. Mineral wool is a good insulator because it holds air still in the matrix of the wool so that heat is not lost. Since air is a poor conductor and a good insulator, this helps to keep energy within the building. Polystyrene is also a good insulator and is able to keep cool things cool and hot things hot. It has the added advantage of being resistant to moisture, mould and mildew.

An **insulator** is a material that does not allow a transfer of electricity or energy. Materials that are poor thermal conductors can also be described as being good thermal insulators.

Investigation: *A closer look at thermal conductivity*

Look at the table below, which shows the thermal conductivity of a number of different materials, and then answer the questions that follow. The higher the number in the second column, the better the material is at conducting heat (i.e. it is a good thermal conductor). Remember that a material that conducts heat efficiently, will also lose heat more quickly than an insulating material.

Material	**Thermal Conductivity ($W \cdot m^{-1} \cdot K^{-1}$)**
Silver	429
Stainless steel	16
Standard glass	1.05
Concrete	0.9 - 2
Red brick	0.69
Water	0.58
Polyethylene (plastic)	0.42 - 0.51
Wood	0.04 - 0.12
Polystyrene	0.03
Air	0.024

Use this information to answer the following questions:

1. Name two materials that are good thermal conductors.
2. Name two materials that are good insulators.
3. Explain why:
   a. Red brick is a better choice than concrete for building houses that need less internal heating.
   b. Stainless steel is good for making cooking pots

Magnetic and Non-magnetic Materials

ESAAJ

We have now looked at a number of ways in which matter can be grouped, such as into metals, semi-metals and non-metals; electrical conductors and insulators, and thermal conductors and insulators. One way in which we can further group metals, is to divide them into those that are **magnetic** and those that are **non-magnetic.**

See video: VPaga at www.everythingscience.co.za

DEFINITION: Magnetism

Magnetism is a force that certain kinds of objects, which are called 'magnetic' objects, can exert on each other without physically touching. A magnetic object is surrounded by a magnetic 'field' that gets weaker as one moves further away from the object.

A metal is said to be **ferromagnetic** if it can be magnetised (i.e. made into a magnet). If you hold a magnet very close to a metal object, it may happen that its own electrical field will be induced and the object becomes magnetic. Some metals keep their magnetism for longer than others. Look at iron and steel for example. Iron loses its magnetism quite quickly if it is taken away from the magnet. Steel on the other hand will stay magnetic for a longer time. Steel is often used to make permanent magnets that can be used for a variety of purposes.

Magnet

Photo by Aney on Wikimedia

Magnets are used to sort the metals in a scrap yard, in compasses to find direction, in the magnetic strips of video tapes and ATM cards where information must be stored, in computers and TV's, as well as in generators and electric motors.

Investigation: *Magnetism*

You can test whether an object is magnetic or not by holding another magnet close to it. If the object is attracted to the magnet, then it too is magnetic.

Find five objects in your classroom or your home and test whether they are magnetic or not. Then complete the table below:

Object	Magnetic or non-magnetic

Group Discussion: Properties of materials

In groups of 4-5, discuss how knowledge of the properties of materials has allowed::

- society to develop advanced computer technology
- homes to be provided with electricity
- society to find ways to conserve energy
- indigenous peoples to cook their food

Chapter 2 | Summary

See the summary presentation (Presentation: VPcyl at www.everythingscience.co.za)

- All the objects and substances that we see in the world are made of **matter**.
- This matter can be classified according to whether it is a **mixture** or a **pure substance**.

- A **mixture** is a combination of two or more substances, where these substances are not bonded (or joined) to each other and no chemical reaction occurs between the substances. Examples of mixtures are air (a mixture of different gases) and cereal in milk.
- The main **characteristics** of mixtures are that the substances that make them up are not in a fixed ratio, these substances keep their physical properties and these substances can be separated from each other using mechanical means.

- A **heterogeneous mixture** is one that consists of two or more substances. It is non-uniform and the different components of the mixture can be seen. An example would be a mixture of sand and water.
- A **homogeneous mixture** is one that is uniform, and where the different components of the mixture cannot be seen. An example would be salt in water.
- Pure substances can be further divided into **elements** and **compounds**.
- An **element** is a substance that cannot be broken down into other substances through chemical means.
- All the elements are found on the **periodic table**. Each element has its own chemical symbol. Examples are iron (Fe), sulphur (S), calcium (Ca), magnesium (Mg) and fluorine (F).
- A **compound** is a A substance made up of two or more different elements that are joined together in a fixed ratio. Examples of compounds are sodium chloride (NaCl), iron sulphide (FeS), calcium carbonate ($CaCO_3$) and water (H_2O).
- When **naming compounds** and writing their **chemical formula**, it is important to know the elements that are in the compound, how many atoms of each of these elements will combine in the compound and where the elements are in the periodic table. A number of rules can then be followed to name the compound.
- Another way of classifying matter is into **metals** (e.g. iron, gold, copper), **metalloids** (e.g. silicon and germanium) and **non-metals** (e.g. sulphur, phosphorus and nitrogen).

- **Metals** are good electrical and thermal conductors, they have a shiny lustre, they are malleable and ductile, and they have a high melting point. Metals also have a high density. These properties make metals very useful in electrical wires, cooking utensils, jewellery and many other applications.
- Matter can also be classified into **electrical conductors**, **semi-conductors** and **insulators**.
- An **electrical conductor** allows an electrical current to pass through it. Most metals are good electrical conductors.
- An **electrical insulator** is a non-conducting material that does not carry any charge. Examples are plastic, wood, cotton material and ceramic.
- Materials may also be classified as **thermal conductors** or **thermal insulators** depending on whether or not they are able to conduct heat.
- Materials may also be **magnetic** or **non-magnetic**. Magnetism is a force that certain kinds of objects, which are called 'magnetic' objects, can exert on each other without physically touching. A magnetic object is surrounded by a magnetic 'field' that gets weaker as one moves further away from the object.

Chapter 2 End of chapter exercises

1. Which of the following can be classified as a mixture:
   a. sugar
   b. table salt
   c. air
   d. iron
2. An element can be defined as:
   a. A substance that cannot be separated into two or more substances by ordinary chemical (or physical) means
   b. A substance with constant composition
   c. A substance that contains two or more substances, in definite proportion by weight
   d. A uniform substance
3. Classify each of the following substances as an *element*, a *compound*, a *homogeneous mixture*, or a *heterogeneous mixture*: salt, pure water, soil, salt water, pure air, carbon dioxide, gold and bronze.

4. Look at the table below. In the first column (A) is a list of substances. In the second column (B) is a description of the group that each of these substances belongs in. Match up the *substance* in Column A with the *description* in Column B.

Column A	**Column B**
1. iron	**A.** a compound containing 2 elements
2. H_2S	**B.** a heterogeneous mixture
3. sugar solution	**C.** a metal alloy
4. sand and stones	**D.** an element
5. steel	**E.** a homogeneous mixture

5. You are given a test tube that contains a mixture of iron filings and sulphur. You are asked to weigh the amount of iron in the sample.
   a. Suggest one method that you could use to separate the iron filings from the sulphur.
   b. What property of metals allows you to do this?

6. Given the following descriptions, write the chemical formula for each of the following substances:

  a. silver metal
  b. a compound that contains only potassium and bromine
  c. a gas that contains the elements carbon and oxygen in a ratio of 1:2

7. Give the names of each of the following compounds:

  a. NaBr
  b. $Ba(NO_2)_2$
  c. SO_2
  d. H_2SO_4

8. Give the formula for each of the following compounds:

  a. iron (II) sulphate
  b. boron trifluoride
  c. potassium permanganate
  d. zinc chloride

9. For each of the following materials, say what properties of the material make it important in carrying out its particular function.

  a. **tar** on roads
  b. **iron** burglar bars
  c. **plastic** furniture
  d. **metal** jewellery
  e. **clay** for building
  f. **cotton** clothing

More practice video solutions or help at www.everythingscience.co.za

(1.) 0006 (2.) 0007 (3.) 0008 (4.) 0009 (5.) 000a (6.) 000b (7.) 000c (8.) 000d
(9.) 000e

States of matter and the kinetic molecular theory 3

States of matter ESAAK

In this chapter we will explore the states of matter and then look at the kinetic molecular theory. Matter exists in three states: solid, liquid and gas. We will also examine how the kinetic theory of matter helps explain boiling and melting points as well as other properties of matter.
See introductory video: (Video: VPajx at www.everythingscience.co.za)

All matter is made up of particles. We can see this when we look at diffusion.

> **DEFINITION: Diffusion**
>
> Diffusion is the movement of particles from a high concentration to a low concentration.

Diffusion can be seen as a spreading out of particles resulting in an even distribution of the particles. You can see diffusion when you place a drop of food colouring in water. The colour slowly spreads out through the water. If matter were not made of particles that are constantly moving then we would only see a clump of colour when we put the food colouring in water, as there would be nothing that could move about and mix in with the water.

Food colouring in water

Picture by LadyDayDream on Flickr.com

Diffusion is a result of the constant thermal motion of particles. In 1828 Robert Brown observed that pollen grains suspended in water moved about in a rapid, irregular motion. This motion has since become known as **Brownian motion**. Brownian motion is essentially diffusion of many particles. Brownian motion can also be seen as the random to and fro movement of particles.

Matter exists in one of three states, namely solid, liquid and gas. A solid has a fixed shape and volume. A liquid takes on the shape of the container that it is in. A gas completely fills the container that it is in. Matter can change between these states by either adding heat or removing heat. This is known as a **change of state**. As we heat an object (e.g. water) it goes from a solid to a liquid to a gas. As we cool an object it goes from a gas to a liquid to a solid. The changes of state that you should know are:

- **Melting**

> *DEFINITION: Melting point*
>
> The temperature at which a *solid* changes its phase or state to become a *liquid*. The process is called melting.

- **Freezing**

> *DEFINITION: Freezing point*
>
> The temperature at which a *liquid* changes its phase to become a *solid*. The process is called freezing.

- **Evaporation**
  Evaporation is the process of going from a liquid to a gas. Evaporation from a liquid's surface can happen at a wide range of temperatures. If more energy is added then bubbles of gas appear inside the liquid and this is known as boiling.

> *DEFINITION: Boiling point*
>
> The temperature at which a *liquid* changes its phase to become a *gas*. The process is called evaporation

- **Condensation** is the process of going from gas to liquid.
- **Sublimation** is the process of going from a solid to a gas. The reverse process is called deposition.

If we know the melting and boiling point of a substance then we can say what state (solid, liquid or gas) it will be in at any temperature.

The figure 3.1 summarises these processes:

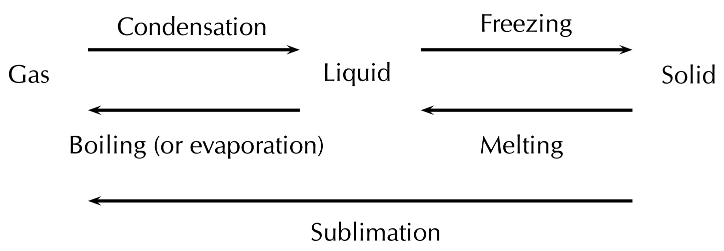


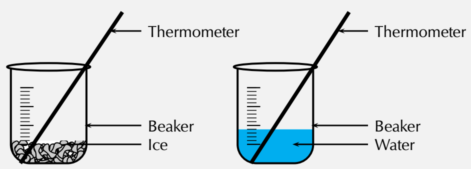
Figure 3.1: Changes in phase

Formal experiment: *Heating and cooling curve of water*

Aim: To investigate the heating and cooling curve of water.

Apparatus:

- beakers
- ice
- Bunsen burner
- thermometer
- water



Method:

1. Place some ice in a beaker.
2. Measure the temperature of the ice and record it.
3. After 1 minute measure the temperature again and record it. Repeat every minute, until at least 3 minutes after the ice has melted.
4. Plot a graph of time versus temperature for the heating of ice.
5. Heat some water in a beaker until it boils. Measure and record the temperature of the water.
6. Remove the water from the heat and measure the temperature every 1 minute, until the beaker is cool to touch.

Warning:
Be careful when handling the beaker of hot water. Do not touch the beaker with your hands, you will burn yourself.

Results:

1. Record your results in the following table:

Heating of ice		Cooling of boiling water	
Time (min)	**Temperature (in °C)**	**Time (min)**	**Temperature (in °C)**
0		0	
1		1	
2		2	
etc.		etc.	

2. Plot a graph of time (independent variable, x-axis) against temperature (dependent variable, y-axis) for the ice melting and the boiling water cooling.

Discussion and conclusion: You should find that the temperature of the ice increases until the first drops of liquid appear and then the temperature remains the same, until all the ice is melted. You should also find that when you cool water down from boiling, the temperature remains constant for a while, then starts decreasing.

In the above experiment, you investigated the heating and cooling curves of water. We can draw heating and cooling curves for any substance. A **heating curve** of a substance gives the changes in temperature as we move from a *solid* to a *liquid* to a *gas*. A **cooling curve** gives the changes in temperature as we move from *gas* to *liquid* to *solid*. An important observation is that as a substance melts or boils, the temperature remains constant until the substance has changed state. This is because all the heat energy goes into breaking or forming the bonds between the molecules.
The following diagram gives an example of what heating and cooling curves look like:

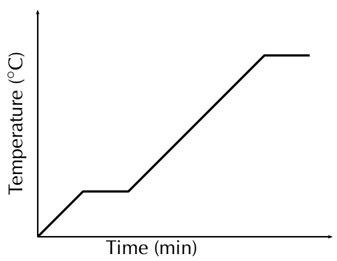


Figure 3.2: Heating curve

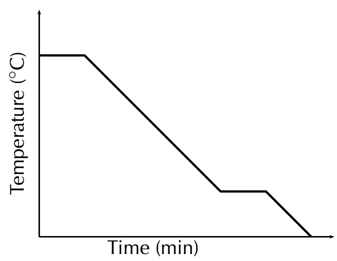


Figure 3.3: Cooling curve

The kinetic molecular theory

ESAAL

The **kinetic theory of matter** helps us to explain why matter exists in different *phases* (i.e. solid, liquid and gas), and how matter can change from one phase to the next. The kinetic theory of matter also helps us to understand other properties of matter. It is important to realise that what we will go on to describe is only a *theory*. It cannot be proved beyond doubt, but the fact that it helps us to explain our observations of changes in phase, and other properties of matter, suggests that it probably is more than just a theory.

Broadly, the kinetic theory of matter says that all matter is composed of **particles** which have a certain amount of **energy** which allows them to move at different speeds depending on the temperature (energy). There are **spaces** between the particles and also **attractive forces** between particles when they come close together.

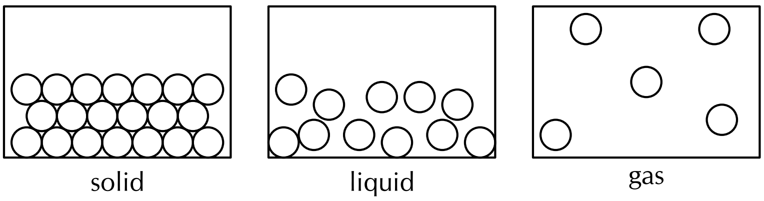


Figure 3.4: The three states of matter

Table 3.2 summarises the characteristics of the particles that are in each phase of matter.

Taking copper as an example we find that in the solid phase the copper atoms have little energy. They vibrate in fixed positions. The atoms are held closely together in a regular pattern called a *lattice*. If the copper is heated, the energy of the atoms increases. This means that some of the copper atoms are able to overcome the forces that are holding them together, and they move away from each other to form *liquid copper*. This is why liquid copper is able to flow, because the atoms are more free to move than when they were in the solid lattice. If the liquid is heated further, it will become a gas. Gas particles have lots of energy and are far away from each other. That is why it is difficult to keep a gas in a specific area! The attractive forces between the particles are very weak. Gas atoms will fill the container they are in. Figure 3.1 shows the changes in phase that may occur in matter, and the names that describe these processes.

Activity:	*The three phases of water*

Property of matter	**Solid**	**Liquid**	**Gas**
Particles	Atoms or molecules	Atoms or molecules	Atoms or molecules
Energy and movement of particles	Low energy - particles vibrate around a fixed point.	Particles have more energy than in the solid phase but less than in the gas phase.	Particles have high energy and are constantly moving.
Spaces between particles	Very little space between particles. Particles are tightly packed together.	Bigger spaces than in solids but smaller than in gases.	Large spaces because of high energy.
Attractive forces between particles.	Very strong forces. Solids have a fixed volume.	Weaker forces than in solids, but stronger forces than in gases.	Weak forces because of the large distance between particles.
Changes in phase.	Solids become liquids or gases if their temperature is increased.	A liquid becomes a gas if its temperature is increased. A liquid becomes a solid if its temperature decreases.	In general a gas becomes a liquid or solid when it is cooled. Particles have less energy and therefore move closer together so that the attractive forces become stronger, and the gas becomes a liquid or a solid.

Water can be in the form of steam, water liquid or ice. Use marbles (or playdough or clay) to represent water molecules. Arrange the marbles to show the three phases of water. Discuss the properties of each of the phases and the processes and energy in changing from the one phase to the other.

Picture by stevendepolo on Flickr.com

Picture by Alan Vernon on Flickr.com

Chapter 3 | Summary

See the summary presentation (Presentation: VPdgh at www.everythingscience.co.za)

- There are three states of matter: solid, liquid and gas.
- **Diffusion** is the movement of particles from a high concentration to a low concentration. Brownian motion is the diffusion of many particles.
- **Melting point** is the temperature at which a *solid* changes its phase to become a *liquid*. The process is called melting.
- **Freezing point** is the temperature at which a liquid changes its phase to become a solid. The process is called freezing.
- Evaporation is the process of going from a liquid to a gas. Evaporation from a liquid's surface can happen at a wide range of temperatures.
- **Boiling point** is the temperature at which a liquid changes phase to become a gas. The process is called evaporation. The reverse process (change in phase from gas to liquid) is called **condensing**.
- Sublimation is the process of going from a solid to a gas.
- The **kinetic theory of matter** attempts to explain the behaviour of matter in different phases.
- The kinetic theory of matter says that all matter is composed of **particles** which have a certain amount of **energy** which allows them to **move** at different speeds depending on the temperature (energy). There are **spaces** between the particles and also **attractive forces** between particles when they come close together.

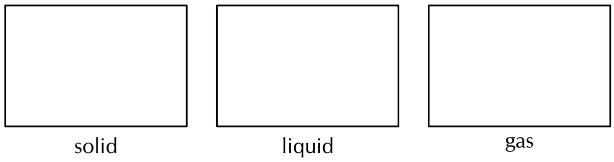
Chapter 3 End of chapter exercises

1. Give one word or term for each of the following descriptions.
   a. The change in phase from a solid to a gas.
   b. The change in phase from liquid to gas.
2. Water has a boiling point of $100°$ C
   a. Define boiling point.
   b. What change in phase takes place when a liquid reaches its boiling point?
3. Describe a solid in terms of the kinetic molecular theory.
4. Refer to the table below which gives the melting and boiling points of a number of elements and then answer the questions that follow. (*Data from http://www.chemicalelements.com*)

Element	**Melting point (°C)**	**Boiling point (°C)**
copper	1083	2567
magnesium	650	1107
oxygen	-218,4	-183
carbon	3500	4827
helium	-272	-268,6
sulphur	112,8	444,6

a. What state of matter (i.e. solid, liquid or gas) will each of these elements be in at room temperature ($25°C$)?
b. Which of these elements has the strongest forces between its atoms? Give a reason for your answer.
c. Which of these elements has the weakest forces between its atoms? Give a reason for your answer.

5. Complete the following submicroscopic diagrams to show what magnesium will look like in the solid, liquid and gas phase.



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(1.) 000f (2.) 000g (3.) 000h (4.) 000i (5.) 000j

The atom 4

Introduction ESAAM

We have now looked at many examples of the types of matter and materials that exist around us and we have investigated some of the ways that materials are classified. But what is it that makes up these materials? And what makes one material different from another? In order to understand this, we need to take a closer look at the building blocks of matter - the **atom**. Atoms are the basis of all the structures and organisms in the universe. The planets, sun, grass, trees, air we breathe and people are all made up of different combinations of atoms.

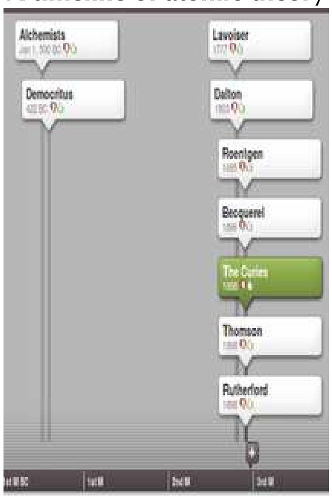
See introductory video: (Video: VPrkk at www.everythingscience.co.za)

Project: **Library assignment: Models of the atom**

Our current understanding of the atom came about over a long period of time, with many different people playing a role. Conduct some research into the development of at least five different ideas of the atom and the people who contributed to it.

Some suggested people to look at are: JJ Thomson, Ernest Rutherford, Marie Curie, JC Maxwell, Max Planck, Albert Einstein, Niels Bohr, Lucretius, LV de Broglie, CJ Davisson, LH Germer, Chadwick, Werner Heisenberg, Max Born, Erwin Schrodinger, John Dalton, Empedocles, Leucippus, Democritus, Epicurus, Zosimos, Maria the Jewess, Geber, Rhazes, Robert Boyle, Henry Cavendish, A Lavoisier and H Becquerel. You do not need to find information on all these people, but try to find information about as many of them as possible. Make a list of five key contributions to a model of the atom and then make a timeline of this information. (You can use an online tool such as Dipity (http://www.dipity.com/) to make a timeline.) Try to get a feel for how it all eventually fit together into the modern understanding of the atom.

A timeline of atomic theory



Models of the atom

ESAAN

It is important to realise that a lot of what we know about the structure of atoms has been developed over a long period of time. This is often how scientific knowledge develops, with one person building on the ideas of someone else. We are going to look at how our modern understanding of the atom has evolved over time.
See video: VPakv at www.everythingscience.co.za

The idea of atoms was invented by two Greek philosophers, Democritus and Leucippus in the fifth century BC. The Greek word $\alpha\tau o\mu o\nu$ (atom) means **indivisible** because they believed that atoms could not be broken into smaller pieces.
Nowadays, we know that atoms are made up of a **positively charged nucleus** in the centre surrounded by **negatively charged electrons**. However, in the past, before the structure of the atom was properly understood, scientists came up with lots of different **models** or **pictures** to describe what atoms look like.

DEFINITION: Model

A model is a representation of a system in the real world. Models help us to understand systems and their properties.

For example, an *atomic model* represents what the structure of an atom *could* look like, based on what we know about how atoms behave. It is not necessarily a true picture of the exact structure of an atom.
Models are often simplified. The small toy cars that you may have played with as a child are models. They give you a good idea of what a real car looks like, but they are much smaller and much simpler. A model cannot always be absolutely accurate and it is important that we realise this, so that we do not build up an incorrect idea about something.

Dalton's model of the atom ESAAO

John Dalton proposed that all matter is composed of very small things which he called atoms. This was not a completely new concept as the ancient Greeks (notably Democritus) had proposed that all matter is composed of small, indivisible (cannot be divided) objects. When Dalton proposed his model electrons and the nucleus were unknown.

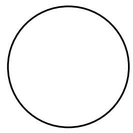


Figure 4.1: The atom according to Dalton

Thomson's model of the atom ESAAP

After the electron was discovered by J.J. Thomson in 1897, people realised that atoms were made up of even smaller particles than they had previously thought. However, the atomic nucleus had not been discovered yet and so the "plum pudding model" was put forward in 1904. In this model, the atom is made up of negative electrons that float in a "soup" of positive charge, much like plums in a pudding or raisins in a fruit cake (figure 4.2). In 1906, Thomson was awarded the Nobel Prize for his work in this field. However, even with the Plum Pudding Model, there was still no understanding of how these electrons in the atom were arranged.

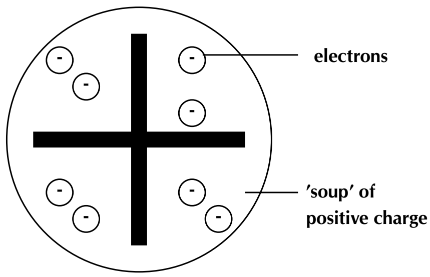


Figure 4.2: The atom according to the Plum Pudding model

FACT

Two other models proposed for the atom were the cubic model and the Saturnian model. In the cubic model, the electrons were imagined to lie at the corners of a cube. In the Saturnian model, the electrons were imagined to orbit a very big, heavy nucleus.

The discovery of **radiation** was the next step along the path to building an accurate picture of atomic structure. In the early twentieth century, Marie and Pierre Curie, discovered that some elements (the *radioactive* elements) emit particles, which are able to pass through matter in a similar way to X-rays (read more about this in Grade 11). It was Ernest Rutherford who, in 1911, used this discovery to revise the model of the atom.

Rutherford's model of the atom ESAAQ

Rutherford carried out some experiments which led to a change in ideas around the atom. His new model described the atom as a tiny, dense, positively charged core called a nucleus surrounded by lighter, negatively charged electrons. Another way of thinking about this model was that the atom was seen to be like a mini solar system where the electrons orbit the nucleus like planets orbiting around the sun. A simplified picture of this is shown alongside. This model is sometimes known as the planetary model of the atom.

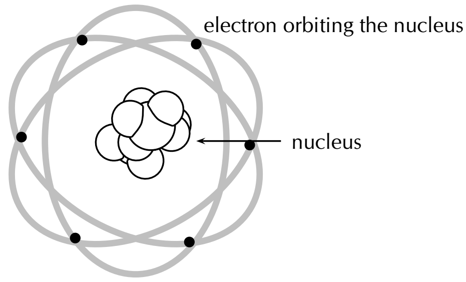


Figure 4.3: Rutherford's model of the atom

Bohr's model of the atom ESAAR

There were, however, some problems with Rutherford's model: for example it could not explain the very interesting observation that atoms only emit light at certain wavelengths or frequencies. Niels Bohr solved this problem by proposing that the electrons could only orbit the nucleus in certain special orbits at different energy levels around the nucleus.

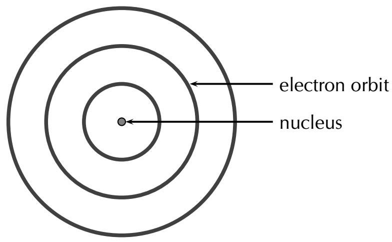


Figure 4.4: Bohr's model of the atom

James Chadwick ESAAS

Rutherford predicted (in 1920) that another kind of particle must be present in the nucleus along with the proton. He predicted this because if there were only positively charged protons in the nucleus, then it should break into bits because of the repulsive forces between the like-charged protons! To make sure that the atom stays electrically neutral, this particle would have to be neutral itself. In 1932 James Chadwick discovered the neutron and measured its mass.

Other models of the atom

ESAAT

Although the most commonly used model of the atom is the Bohr model, scientists are still developing new and improved theories on what the atom looks like. One of the most important contributions to atomic theory (the field of science that looks at atoms) was the development of quantum theory. Schrodinger, Heisenberg, Born and many others have had a role in developing quantum theory.

Exercise 4 - 1

Match the information in column A, with the key discoverer in column B.

Column A	Column B
1. Discovery of electrons and the plum pudding model	**A.** Niels Bohr
2. Arrangement of electrons	**B.** Marie and Pierre Curie
3. Atoms as the smallest building block of matter	**C.** Ancient Greeks and Dalton
4. Discovery of the nucleus	**D.** JJ Thomson
5. Discovery of radiation	**E.** Rutherford

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(1.) 000k

Atomic mass and diameter

ESAAU

It is difficult sometimes to imagine the size of an atom, or its mass, because we cannot see an atom and also because we are not used to working with such small measurements.

How heavy is an atom? ESAAV

It is possible to determine the mass of a single atom in kilograms. But to do this, you would need special instruments and the values you would get would be very clumsy and difficult to work with. The mass of a carbon atom, for example, is about $1,99 \times 10^{-26}$ kg, while the mass of an atom of hydrogen is about $1,67 \times 10^{-27}$ kg. Looking at these very small numbers makes it difficult to compare how much bigger the mass of one atom is when compared to another.

To make the situation simpler, scientists use a different unit of mass when they are describing the mass of an atom. This unit is called the **atomic mass unit** (amu). We can abbreviate (shorten) this unit to just u. Scientists use the **carbon standard** to determine amu. The carbon standard gives carbon an atomic mass of $12,0$ u. Compared to carbon the mass of a hydrogen atom will be 1 u. Atomic mass units are therefore not giving us the *actual* mass of an atom, but rather its mass *relative* to the mass of one (carefully chosen) atom in the periodic table. In other words it is only a number in comparison to another number. The atomic masses of some elements are shown in table 4.1.

Element	**Atomic mass (u)**
Carbon (C)	$12,0$
Nitrogen (N)	$14,0$
Bromine (Br)	$79,9$
Magnesium (Mg)	$24,3$
Potassium (K)	$39,1$
Calcium (Ca)	$40,1$
Oxygen (O)	$16,0$

Table 4.1: The atomic mass number of some of the elements.

The actual value of 1 atomic mass unit is $1,67 \times 10^{-24}$ g or $1,67 \times 10^{-27}$ kg. This is a very tiny mass! If we write it out it looks like this: $0,000\ 000\ 000\ 000\ 000\ 000\ 000\ 000\ 167$ kg. An atom is therefore very very small.

Rutherford's alpha-particle scattering experiment ESAAW

Radioactive elements emit different types of particles. Some of these are positively charged alpha (α) particles. Rutherford wanted to find out where the positive charge in an atom is. He carried out a series of experiments where he bombarded sheets of gold foil with alpha particles (since these would be repelled by the positive nucleus). A simplified diagram of his experiment is shown in figure 4.5.

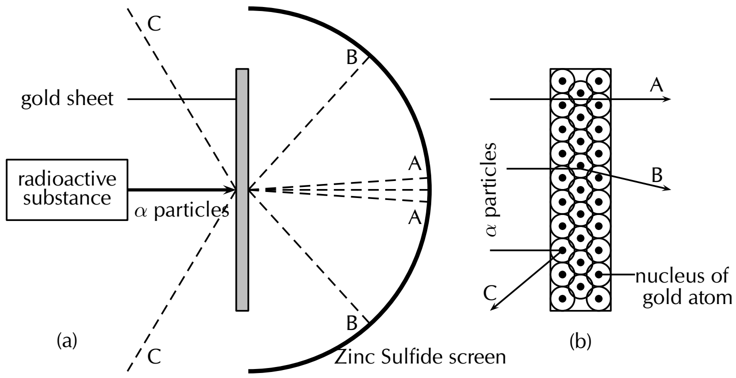


Figure 4.5: Rutherford's gold foil experiment. Figure (a) shows the path of the α particles after they hit the gold sheet. Figure (b) shows the arrangement of atoms in the gold sheets and the path of the α particles in relation to this.

See video: VPald at www.everythingscience.co.za

Rutherford set up his experiment so that a beam of alpha particles was directed at the gold sheets. Behind the gold sheets was a screen made of zinc sulphide. This screen allowed Rutherford to see where the alpha particles were landing. Rutherford knew that the *electrons* in the gold atoms would not really affect the path of the alpha particles, because the mass of an electron is so much smaller than that of a proton. He reasoned that the positively charged *protons* would be the ones to *repel* the positively charged alpha particles and alter their path.

If Thomson's model of the atom was correct then Rutherford would have observed mostly path C in figure 4.5. (C represents alpha particles that are reflected by the positive nucleus). What he found instead was that most of the alpha particles passed through the foil undisturbed and could be detected on the screen directly behind the foil (path A). Some of the particles ended up being slightly deflected onto other parts of the screen (path B). The fact that most particles passed straight through suggested that the positive charge was concentrated in one part of the atom only.

Through this experiment he concluded that the nucleus of the atom is positively charged and situated in the middle of the atom.

How small is the nucleus? ESAAX

The nucleus of an atom is very small. We can compare an atom with a soccer stadium. If the whole atom is as big as a soccer stadium, the nucleus is only the size of a pea in the middle of the stadium! This should help you see that the atom is mainly made up of empty space. If you removed all the empty space from the atoms in your body, you would become the size of a grain of salt!

A stadium

Picture by Shine 2010 on Flickr.com

Relative atomic mass ESAAY

DEFINITION: Relative atomic mass

The relative atomic mass of an element is the average mass of all the naturally occurring isotopes of that element. The units for relative atomic mass are atomic mass units.

The relative atomic mass of an element is the number you will find on the periodic table.

Structure of the atom ESAAZ

As a result of the work done by previous scientists on atomic models, scientists now have a good idea of what an atom looks like. This knowledge is important because it helps us to understand why materials have different properties and why some materials bond with others. Let us now take a closer look at the microscopic structure of the atom (what the atom looks like inside).

See video: VPaln at www.everythingscience.co.za

So far, we have discussed that atoms are made up of a positively charged **nucleus** surrounded by one or more negatively charged **electrons**. These electrons orbit the nucleus. Before we look at some useful concepts we first need to understand what electrons, protons and neutrons are.

The Electron

ESABA

The electron is a very tiny particle. It has a mass of $9,11 \times 10^{-31}$ kg. The electron carries one unit of negative electric charge (i.e. $-1,6 \times 10^{-19}$ C).

The Nucleus

ESABB

Unlike the electron, the nucleus **can** be broken up into smaller building blocks called **protons** and **neutrons**. Together, the protons and neutrons are called **nucleons**.

FACT

Scientists believe that the electron can be treated as a **point particle** or **elementary particle** meaning that it cannot be broken down into anything smaller.

The Proton

The electron carries one unit of **negative** electric charge (i.e. $-1,6 \times 10^{-19}$ C, C is Coulombs). Each proton carries one unit of **positive** electric charge (i.e. $+1,6 \times 10^{-19}$ C). Since we know that atoms are **electrically neutral**, i.e. do not carry any extra charge, then the number of protons in an atom has to be the same as the number of electrons to balance out the positive and negative charge to zero. The total positive charge of a nucleus is equal to the number of protons in the nucleus. The proton is much heavier than the electron ($10\ 000$ times heavier!) and has a mass of $1,6726 \times 10^{-27}$ kg. When we talk about the atomic mass of an atom, we are mostly referring to the combined mass of the protons and neutrons, i.e. the nucleons.

The Neutron

The neutron is electrically neutral i.e. it carries no charge at all. Like the proton, it is much heavier than the electron and its mass is $1,6749 \times 10^{-27}$ kg (slightly heavier than the proton).

	proton	neutron	electron
Mass (kg)	$1,6726 \times 10^{-27}$	$1,6749 \times 10^{-27}$	$9,11 \times 10^{-31}$
Units of charge	$+1$	0	-1
Charge (C)	$1,6 \times 10^{-19}$	0	$-1,6 \times 10^{-19}$

Table 4.2: Summary of the particles inside the atom

Atomic number and atomic mass number ESABC

The chemical properties of an element are determined by the charge of its nucleus, i.e. by the **number of protons**. This number is called the **atomic number** and is denoted by the letter **Z**.

> **FACT**
>
> Currently element 118 is the highest atomic number for an element. Elements of high atomic numbers (from about 93 to 118) do not exist for long as they break apart within seconds of being formed. Scientists believe that after element 118 there may be an "island of stability" in which elements of higher atomic number occur that do not break apart within seconds.

DEFINITION: Atomic number (Z)

The number of protons in an atom

You can find the atomic number on the periodic table (see periodic table at front of book). The atomic number is an integer and ranges from 1 to about 118.

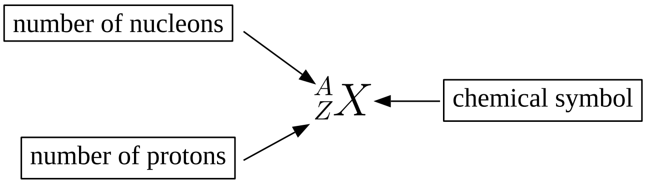
The mass of an atom depends on how many nucleons its nucleus contains. The number of nucleons, i.e. the total number of protons **plus** neutrons, is called the **atomic mass number** and is denoted by the letter **A**.

DEFINITION: Atomic mass number (A)

The number of protons and neutrons in the nucleus of an atom

The atomic number (Z) and the mass number (A) are indicated using a standard notation, for example carbon will look like this: $^{12}_{6}C$

Standard notation shows the chemical symbol, the atomic mass number and the atomic number of an element as follows:



For example, the iron nucleus which has 26 protons and 30 neutrons, is denoted as:

$$^{56}_{26}\text{Fe}$$

where the atomic number is $Z = 26$ and the mass number $A = 56$. The number of neutrons is simply the difference $N = A - Z = 30$.

For a neutral atom the number of electrons is the same as the number of protons, since the charge on the atom must balance. But what happens if an atom gains or loses electrons? Does it mean that the atom will still be part of the same element? A change in the number of electrons of an atom does not change the type of atom that it is. However, the charge of the atom will change. The *neutrality* of the atom has changed. If electrons are *added*, then the atom will become more *negative*. If electrons are *taken away* then the atom will become more *positive*. The atom that is formed in either of these two cases is called an **ion**. An ion is a charged atom. For example: a neutral sodium atom can lose one electron to become a positively charged sodium atom (Na^+). A neutral chlorine atom can gain one electron to become a negatively charged chlorine ion (Cl^-). Another example is Li^+ which has lost one electron and now has only 2 electrons, instead of 3. Or consider F^- which has gained one electron and now has 10 electrons instead of 9.

FACT

A nuclide is a distinct kind of atom or nucleus characterised by the number of protons and neutrons in the atom. To be absolutely correct, when we represent atoms like we do here, then we should call them nuclides.

Tip

Do not confuse the notation we have used here with the way this information appears on the periodic table. On the periodic table, the atomic number usually appears in the top left-hand corner of the block or immediately above the element's symbol. The number below the element's symbol is its **relative atomic mass**. This is not exactly the same as the atomic mass number. This will be explained in "Isotopes". The example of iron is shown below.

26
Fe
55.85

Example 1: *Standard notation*

QUESTION

Use standard notation to represent sodium and give the number of protons, neutrons and electrons in the element.

SOLUTION

Step 1 : ***Give the element symbol***

Na

Step 2 : ***Find the number of protons***

Sodium has 11 protons, so we have: $_{11}Na$

Step 3 : ***Find the number of electrons***

Sodium is neutral, so it has the same number of electrons as protons. The number of electrons is 11.

Step 4 : ***Find A***

From the periodic table we see that $A = 23$.

Step 5 : ***Work out the number of neutrons***

We know A and Z so we can find N: $N = A - Z = 23 - 11 = 12$.

Step 6 : ***Write the answer***

In standard notation sodium is given by: ${}^{23}_{11}Na$. The number of protons is 11, the number of neutrons is 12 and the number of electrons is 11.

Exercise 4 - 2

1. Explain the meaning of each of the following terms:
   a. nucleus
   b. electron
   c. atomic mass
2. Complete the following table:

Element	Atomic mass units	Atomic number	Number of protons	Number of electrons	Number of neutrons
Mg	24	12			
O			8		
		17			
Ni				28	
	40				20
Zn					
					0
C	12			6	
Al^{3+}		13			
O^{2-}				18	

3. Use standard notation to represent the following elements:
   a. potassium
   b. copper
   c. chlorine
4. For the element $^{35}_{17}$Cl, give the number of...
   a. protons
   b. neutrons
   c. electrons

   ... in the atom.

5. Which of the following atoms has 7 electrons?
   a. $^{5}_{2}$He
   b. $^{13}_{6}$C
   c. $^{7}_{3}$Li
   d. $^{15}_{7}$N
6. In each of the following cases, give the number or the element symbol represented by X.
   a. $^{40}_{18}$X
   b. $^{x}_{20}$Ca
   c. $^{31}_{x}$P
7. Complete the following table:

	A	Z	N
$^{235}_{92}$U			
$^{238}_{92}$U			

   In these two different forms of uranium...
   a. What is the *same*?
   b. What is *different*?

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(1.) 000m (2.) 000n (3.) 000p (4.) 000q (5.) 000r (6.) 000s (7.) 000t

Isotopes

ESABD

FACT

In Greek, "same place" reads as ἴσος τόπος (isos topos). This is why atoms which have the same number of protons, but different numbers of neutrons, are called *isotopes*. They are in the same place on the periodic table!

The chemical properties of an element depend on the number of protons and electrons inside the atom. So if a neutron or two is added or removed from the nucleus, then the chemical properties will not change. This means that such an atom would remain in the same place in the periodic table. For example, no matter how many neutrons we add or subtract from a nucleus with 6 protons, that element will **always** be called carbon and have the element symbol C (see the periodic table). Atoms which have the same number of protons (i.e. same atomic number Z), but a different number of neutrons (i.e. different N and therefore different mass number A), are called **isotopes**.

DEFINITION: Isotope

Isotopes of an element have the same number of protons (same Z), but a different number of neutrons (different N).

The chemical properties of the different isotopes of an element are the same, but they might vary in how stable their nucleus is. We can also write elements as E - A where the E is the element symbol and the A is the atomic mass of that element. For example Cl-35 has an atomic mass of 35 u (17 protons and 18 **neutrons**), while Cl-37 has an atomic mass of 37 u (17 protons and 20 **neutrons**).
In nature the different isotopes occur in different percentages. For example Cl-35 might make up 75% of all chlorine atoms on Earth, and Cl-37 makes up the remaining 25%. The following worked example will show you how to calculate the average atomic mass for these two isotopes:

Example 2: *The relative atomic mass of an isotopic element*

QUESTION

The element chlorine has two isotopes, chlorine-35 and chlorine-37. The abundance of these isotopes when they occur naturally is 75% chlorine-35 and 25% chlorine-37.

Calculate the average relative atomic mass for chlorine.

SOLUTION

Step 1 : ***Calculate the mass contribution of chlorine-*35 *to the average relative atomic mass***

75% of the chlorine atoms has a mass of of 35 u

$$\text{Contribution of Cl-35} = (\tfrac{75}{100} \times 35) = 26,25 \text{ u}$$

Step 2 : ***Calculate the contribution of chlorine-*37 *to the average relative atomic mass***

25% of the chlorine atoms has a mass of of 37 u

$$\text{Contribution of Cl-37} = (\tfrac{25}{100} \times 37) = 9,25 \text{ u}$$

Step 3 : ***Add the two values to arrive at the average relative atomic mass of chlorine***

$$\text{Relative atomic mass of chlorine} = 26,25 \text{ u} + 9,25 \text{ u} = 35,5 \text{ u}$$

If you look on the periodic table (see front of book), the average relative atomic mass for chlorine is $35,5$ u. See simulation: (Simulation: VPcyv at www.everythingscience.co.za)

Exercise 4 - 3

1. Atom A has 5 protons and 5 neutrons, and atom B has 6 protons and 5 neutrons. These atoms are:
   - **a**. allotropes
   - **b**. isotopes
   - **c**. isomers
   - **d**. atoms of different elements
2. For the sulphur isotopes, $^{32}_{16}$S and $^{34}_{16}$S, give the number of:
   - **a**. protons
   - **b**. nucleons
   - **c**. electrons
   - **d**. neutrons
3. Which of the following are isotopes of $^{35}_{17}$Cl?
   - **a**. $^{17}_{35}$Cl
   - **b**. $^{35}_{17}$Cl
   - **c**. $^{37}_{17}$Cl

4. Which of the following are isotopes of U-235? (E represents an element symbol)
   a. $^{238}_{92}$E
   b. $^{238}_{90}$E
   c. $^{235}_{92}$E
5. Complete the table below:

Isotope	Z	A	Protons	Neutrons	Electrons
Carbon-12					
Carbon-14					
Iron-54					
Iron-56					
Iron-57					

6. If a sample contains $19,9\%$ boron-10 and $80,1\%$ boron-11, calculate the relative atomic mass of an atom of boron in that sample.
7. If a sample contains 79% Mg-24, 10% Mg-25 and 11% Mg-26, calculate the relative atomic mass of an atom of magnesium in that sample.
8. For the element $^{234}_{92}$U (uranium), use standard notation to describe:
   a. the isotope with 2 fewer neutrons
   b. the isotope with 4 more neutrons
9. Which of the following are isotopes of $^{40}_{20}$Ca?
   a. $^{40}_{19}$K
   b. $^{42}_{20}$Ca
   c. $^{40}_{18}$Ar
10. For the sulphur isotope $^{33}_{16}$S, give the number of:
    a. protons
    b. nucleons
    c. electrons
    d. neutrons

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(1.) 000x (2.) 000y (3.) 000z (4.) 0010 (5.) 0011 (6.) 0012

(7.) 0013 (8.) 0014 (9.) 0015 (10.) 0016

Electronic configuration ESABE

The energy of electrons ESABF

The electrons of an atom all have the same charge and the same mass, but each electron has a different amount of *energy*. Electrons that have the *lowest* energy are found closest to the nucleus (where the attractive force of the positively charged nucleus is the greatest) and the electrons that have *higher* energy (and are able to overcome the attractive force of the nucleus) are found further away.

See video: VPama at www.everythingscience.co.za

Electron arrangement ESABG

We will start with a very simple view of the arrangement or configuration of electrons around an atom. This view simply states that electrons are arranged in energy levels (or shells) around the nucleus of an atom. These energy levels are numbered 1, 2, 3, etc. Electrons that are in the first energy level (energy level 1) are closest to the nucleus and will have the lowest energy. Electrons further away from the nucleus will have a higher energy.

In the following examples, the energy levels are shown as concentric circles around the central nucleus. The important thing to know for these diagrams is that the first energy level can hold 2 electrons, the second energy level can hold 8 electrons and the third energy level can hold 8 electrons.

1. **Lithium**
   Lithium (Li) has an atomic number of 3, meaning that in a neutral atom, the number of electrons will also be 3. The first two electrons are found in the first energy level, while the third electron is found in the second energy level (Figure 4.6).

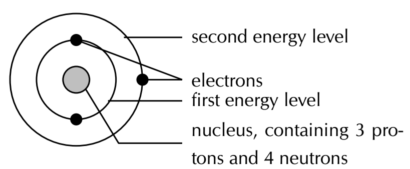


Figure 4.6: Electron arrangement of a lithium atom.

2. **Fluorine**

Fluorine (F) has an atomic number of 9, meaning that a neutral atom also has 9 electrons. The first 2 electrons are found in the first energy level, while the other 7 are found in the second energy level (Figure 4.7).

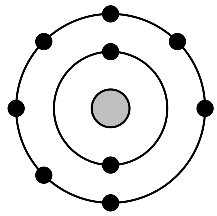


Figure 4.7: Electron arrangement of a fluorine atom.

3. **Neon**

Neon (Ne) has an atomic number of 10, meaning that a neutral atom also has 10 electrons. The first 2 electrons are found in the first energy level and the last 8 are found in the second energy level. (Figure 4.8).

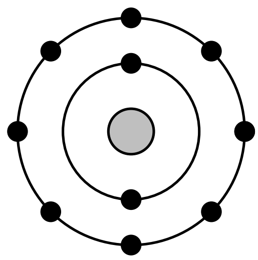


Figure 4.8: Electron arrangement of a neon atom.

But the situation is slightly more complicated than this. Within each energy level, the electrons move in **orbitals**. An orbital defines the spaces or regions where electrons move.

> *DEFINITION: Atomic orbital*
>
> An atomic orbital is the region in which an electron may be found around a single atom.

Note

Each block in figure 4.9 is able to hold two electrons. This means that the s orbital can hold two electrons, while the p orbital can hold a total of six electrons, two in each of the three blocks.

The first energy level contains only one s orbital, the second energy level contains one s orbital and three p orbitals and the third energy level contains one s orbital and three p orbitals (as well as five d orbitals). Within each energy level, the s orbital is at a lower energy than the p orbitals. This arrangement is shown in Figure 4.9.

This diagram also helps us when we are working out the electron configuration of an element. The electron configuration of an element is the arrangement of the electrons in the shells and subshells. There are a few guidelines for working out the electron configuration. These are:

- Each orbital can only hold **two electrons**. Electrons that occur together in an orbital are called an **electron pair**.
- An electron will always try to enter an orbital with the lowest possible energy.

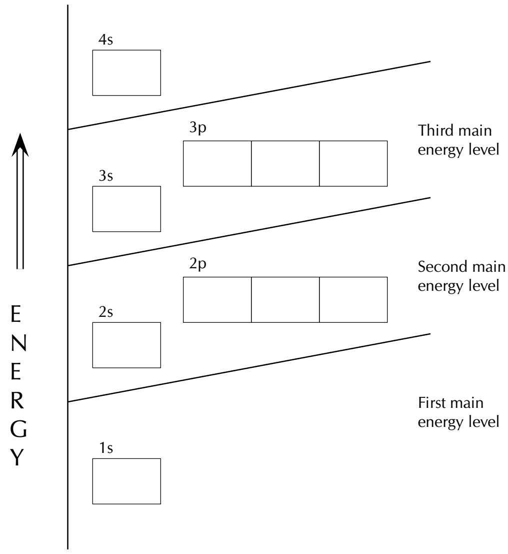


Figure 4.9: The positions of the first ten orbitals of an atom on an energy diagram.

- An electron will occupy an orbital on its own, rather than share an orbital with another electron. An electron would also rather occupy a lower energy orbital *with* another electron, before occupying a higher energy orbital. In other words, within one energy level, electrons will fill an s orbital before starting to fill p orbitals.
- The s subshell can hold 2 electrons
- The p subshell can hold 6 electrons

The way that the electrons are arranged in an atom is called its **electron configuration**.

Tip

When there are two electrons in an orbital, the electrons are called an **electron pair**. If the orbital only has one electron, this electron is said to be an **unpaired electron**. Electron pairs are shown with arrows pointing in opposite directions.

DEFINITION: Electron configuration

Electron configuration is the arrangement of electrons in an atom, molecule or other physical structure.

Aufbau diagrams ESABH

An element's electron configuration can be represented using **Aufbau diagrams** or energy level diagrams. An Aufbau diagram uses arrows to represent electrons. You can use the following steps to help you to draw an Aufbau diagram:

1. Determine the number of electrons that the atom has.
2. Fill the s orbital in the first energy level (the 1s orbital) with the first two electrons.
3. Fill the s orbital in the second energy level (the 2s orbital) with the second two electrons.
4. Put one electron in each of the three p orbitals in the second energy level (the 2p orbitals) and then if there are still electrons remaining, go back and place a second electron in each of the 2p orbitals to complete the electron pairs.
5. Carry on in this way through each of the successive energy levels until all the electrons have been drawn.

FACT

Aufbau is the German word for "building up". Scientists used this term since this is exactly what we are doing when we work out electron configuration, we are building up the atoms structure.

You can think of Aufbau diagrams as being similar to people getting on a bus or a train. People will first sit in empty seats with empty seats between them and the other people (unless they know the people and then they will sit next to them). This is the lowest energy. When all the seats are filled like this, any more people that get on will be forced to sit next to someone. This is higher in energy. As the bus or train fills even more the people have to stand to fit on. This is the highest energy.

Hund's rule and Pauli's principle

Sometimes people refer to Hund's rule for electron configuration. This rule simply says that electrons would rather be in a subshell on their own than share a subshell. This is why when you are filling the subshells you put one electron in each subshell and then go back and fill the subshell, before moving onto the next energy level.

Pauli's exclusion principle simply states that electrons have a property known as spin and that two electrons in a subshell will not spin the same way. This is why we draw electrons as one arrow pointing up and one arrow pointing down.

Spectroscopic electron configuration notation — *ESABI*

A special type of notation is used to show an atom's electron configuration. The notation describes the energy levels, orbitals and the number of electrons in each. For example, the electron configuration of lithium is $1s^2 2s^1$. The number and letter describe the energy level and orbital and the number above the orbital shows how many electrons are in that orbital.

Aufbau diagrams for the elements fluorine and argon are shown in Figures 4.10 and 4.11 respectively. Using spectroscopic notation, the electron configuration of fluorine is $1s^2 2s^2 2p^5$ and the electron configuration of argon is $1s^2 2s^2 2p^5 3s^2 3p^6$.

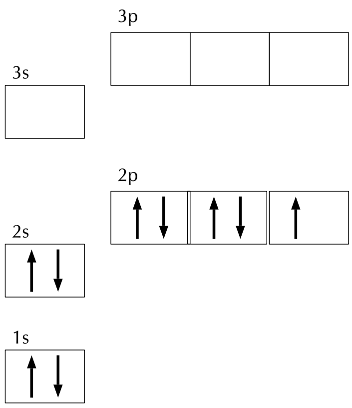


Figure 4.10: An Aufbau diagram showing the electron configuration of fluorine

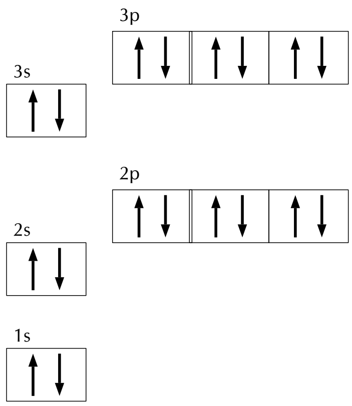


Figure 4.11: An Aufbau diagram showing the electron configuration of argon

Tip

The spectroscopic electron configuration can be written in shorter form. This form is written as [noble gas]electrons, where the noble gas is the nearest one that occurs before the element. For example, magnesium can be represented as [Ne]$3s^2$ and carbon as [H]$2s^2 2p^2$. This is known as the condensed electron configuration.

Example 3: *Aufbau diagrams and spectroscopic electron configuration*

QUESTION

Give the electron configuration for nitrogen (N) and draw an Aufbau diagram.

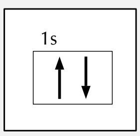
SOLUTION

Step 1 : ***Give the number of electrons***

Nitrogen has seven electrons.

Step 2 : ***Place two electrons in the*** $1s$ ***orbital***

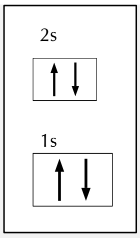
We start by placing two electrons in the 1s orbital: $1s^2$.



Now we have five electrons left to place in orbitals.

Step 3 : ***Place two electrons in the*** $2s$ ***orbital***

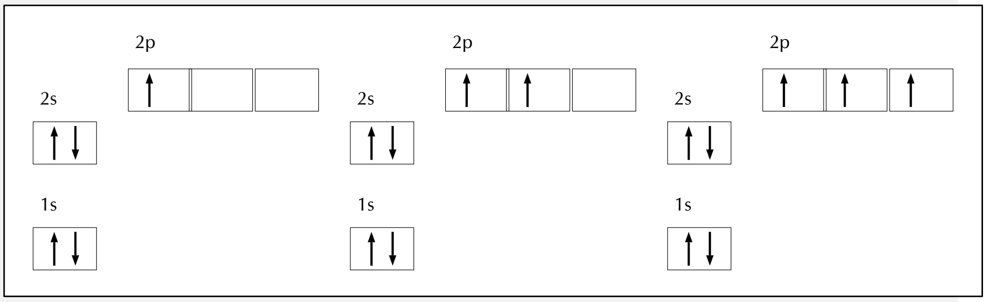
We put two electrons in the 2s orbital: $2s^2$.



There are now three electrons to place in orbitals.

Step 4 : ***Place three electrons in the 2p orbital***

We place three electrons in the 2p orbital: $2p^3$.



Step 5 : ***Write the final answer***

The electron configuration is: $1s^2 2s^2 2p^3$. The Aufbau diagram is given in the step above.

Aufbau diagrams for ions

When a neutral atom loses an electron it becomes positively charged and we call it a cation. For example, sodium will lose one electron and become Na^+ or calcium will lose two electrons and become Ca^{2+}. In each of these cases, the outermost electron(s) will be lost.

When a neutral atom gains an electron it becomes negatively charged and we call it an anion. For example chlorine will gain one electron and become Cl^- or oxygen will gain two electrons and become O^{2-}.

Aufbau diagrams and electron configurations can be done for cations and anions as well. The following worked example will show you how.

Example 4: *Aufbau diagram for an ion*

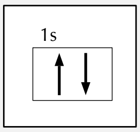
QUESTION

Give the electron configuration for (O^{2-}) and draw an Aufbau diagram.

SOLUTION

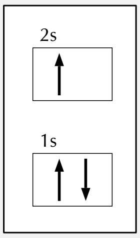
Step 1 : ***Give the number of electrons***
Oxygen has eight electrons. The oxygen anion has gained two electrons and so the total number of electrons is ten.

Step 2 : ***Place two electrons in the 1s orbital***
We start by placing two electrons in the 1s orbital: $1s^2$.



Now we have eight electrons left to place in orbitals.

Step 3 : ***Place two electrons in the 2s orbital***
We put two electrons in the 2s orbital: $2s^2$.



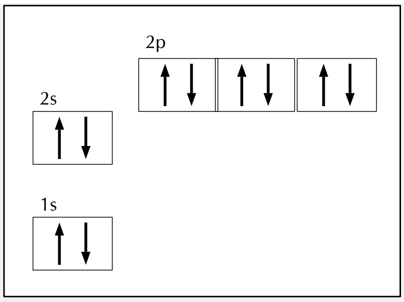
There are now six electrons to place in orbitals.

Step 4 : ***Place six electrons in the 2p orbital***
We place six electrons in the 2p orbital: $2p^6$.

Step 5 : ***Write the final answer***
The electron configuration is: $1s^2 2s^2 2p^6$. The Aufbau diagram is:

FACT

When we draw the orbitals we draw a shape that has a boundary (i.e. a closed shape). This represents the distance from the nucleus in which we are 95% sure that we will find the electrons. In reality the electrons of an atom could be found any distance away from the nucleus.



Orbital shapes ESABJ

Each of the orbitals has a different shape. The s orbitals are spherical and the p orbitals are dumbbell shaped.

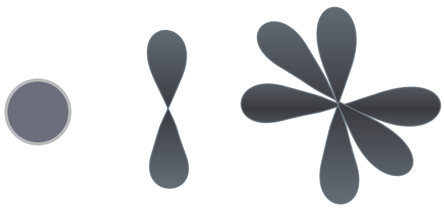


Figure 4.12: The orbital shapes. From left to right: an s orbital, a p orbital, the three p orbitals

Core and valence electrons ESABK

Electrons in the outermost energy level of an atom are called **valence electrons**. The electrons that are in the energy shells closer to the nucleus are called **core electrons**. Core electrons are all the electrons in an atom, excluding the valence electrons. An element that has its valence energy level full is *more stable* and *less likely to react* than other elements with a valence energy level that is not full.

DEFINITION: Valence electrons

The electrons in the outermost energy level of an atom

DEFINITION: Core electrons

All the electrons in an atom, excluding the valence electrons

Exercise 4 - 4

Complete the following table:

Element or Ion	Electron configuration	Core electrons	Valence electrons
Potassium (K)			
Helium (He)			
Oxygen ion (O^{2-})			
Magnesium ion (Mg^{2+})			

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(1.) 0017

The importance of understanding electron configuration ESABL

By this stage, you may well be wondering why it is important for you to understand how electrons are arranged around the nucleus of an atom. Remember that during chemical reactions, when atoms come into contact with one another, it is the *electrons* of these atoms that will interact first. More specifically, it is the **valence electrons** of the atoms that will determine how they react with one another.

To take this a step further, an atom is at its most stable (and therefore *unreactive*) when all its orbitals are full. On the other hand, an atom is least stable (and therefore most *reactive*) when its valence electron orbitals are not full. This will make more sense when we go on to look at chemical bonding in a later chapter. To put it simply, the valence electrons are largely responsible for an element's chemical behaviour and elements that have the same number of valence electrons often have similar chemical properties.

The most stable configurations are the ones that have full energy levels. These configurations occur in the noble gases. The noble gases are very stable elements that do not react easily (if at all) with any other elements. This is due to the full energy levels. All elements would like to reach the most stable electron configurations, i.e. all elements want to be noble gases. This principle of stability is sometimes referred to as the octet rule. An octet is a set of 8, and the number of electrons in a full energy level is 8.

See video: VPamk at www.everythingscience.co.za

Informal experiment: *Flame tests*

Aim: To determine what colour a metal cation will cause a flame to be.

Apparatus:

- Watch glass
- Bunsen burner
- methanol
- tooth picks (or skewer sticks)
- metal salts (e.g. NaCl, $CuCl_2$, $CaCl_2$, KCl, etc.)
- metal powders (e.g. copper, magnesium, zinc, iron, etc.)

Picture by offbeatcin on Flickr.com

Warning:

Be careful when working with Bunsen burners as you can easily burn yourself. Make sure all scarves/loose clothing are securely tucked in and long hair is tied back. Ensure that you work in a well-ventilated space and that there is nothing flammable near the open flame.

Method: For each salt or powder do the following:

1. Dip a clean tooth pick into the methanol
2. Dip the tooth pick into the salt or powder
3. Wave the tooth pick through the flame from the Bunsen burner. DO NOT hold the tooth pick in the flame, but rather wave it back and forth through the flame.
4. Observe what happens

Results: Record your results in a table, listing the metal salt and the colour of the flame.
Conclusion: You should have observed different colours for each of the metal salts and powders that you tested.

The above experiment on flame tests relates to the line emission spectra of the metals. These line emission spectra are a direct result of the arrangement of the electrons in metals. Each metal salt has a uniquely coloured flame.

Exercise 4 - 5

1. Draw Aufbau diagrams to show the electron configuration of each of the following elements:

   a. magnesium
   b. potassium
   c. sulphur
   d. neon
   e. nitrogen

2. Use the Aufbau diagrams you drew to help you complete the following table:

Element	No. of energy levels	No. of electrons	Electron configuration (standard notation)
Mg			
K			
S			
Ne			
N			
Ca^{2+}			
Cl^{-}			

3. Rank the elements used above in order of *increasing reactivity*. Give reasons for the order you give.

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(1.-3.) 0018

Earlier in this chapter, we talked about different "models" of the atom. In science, one of the uses of models is that they can help us to understand the structure of something that we can't see. In the case of the atom, models help us to build a picture in our heads of what the atom looks like.

Group Discussion: Building a model of an atom

In groups of 3-4, you are going to build a model of an atom. Each group will get a different element to represent. Before you start, think about these questions:

- What information do I have about the structure of the atom? (e.g. what parts make it up? how big is it?)
- What materials can I use to represent these parts of the atom as accurately as I can?
- How will I put all these different parts together in my model?

As a group, share your ideas and then plan how you will build your model. Once you have built your model, discuss the following questions:

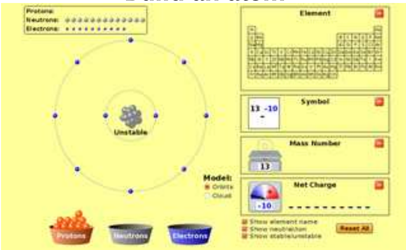
- Does our model give a good idea of what the atom actually looks like?
- In what ways is our model *inaccurate*? For example, we know that electrons *move* around the atom's nucleus, but in your model, it might not have been

possible for you to show this.

- Are there any ways in which our model could be improved?

Now look at what other groups have done. Discuss the same questions for each of the models you see and record your answers.

Build an atom



See simulation: (Simulation: VPcmk at www.everythingscience.co.za)

See simulation: (Simulation: VPcph at www.everythingscience.co.za)

Chapter 4 | Summary

See the summary presentation (Presentation: VPcre at www.everythingscience.co.za)

- Some of the scientists who have contributed to the theory of the atom include **J.J.Thomson** (discovery of the electron, which led to the Plum Pudding Model of the atom), **Marie and Pierre Curie** (work on radiation), **Ernest Rutherford** (discovery that positive charge is concentrated in the centre of the atom) and **Niels Bohr** (the arrangement of electrons around the nucleus in energy levels).
- Because of the very small mass of atoms, their mass is measured in **atomic mass units** (u). $1 \text{ u} = 1,67 \times 10^{-24} \text{ g}$.
- The **relative atomic mass** of an element is the average mass of all the naturally occurring isotopes of that element. The units for relative atomic mass are atomic mass units. The relative atomic mass is written under the elements' symbol on the periodic table.
- An atom is made up of a central **nucleus** (containing **protons** and **neutrons**), surrounded by **electrons**. Most of the atom is empty space.
- The **atomic number** (Z) is the number of protons in an atom.
- The **atomic mass number** (A) is the number of protons and neutrons in the nucleus of an atom.

- The **standard notation** that is used to write an element, is $^{A}_{Z}X$, where X is the element symbol, A is the atomic mass number and Z is the atomic number.
- The **isotope** of a particular element is made up of atoms which have the same number of protons as the atoms in the original element, but a different number of neutrons. This means that not all atoms of an element will have the same atomic mass.
- Within each energy level, an electron may move within a particular shape of **orbital**. An orbital defines the space in which an electron is most likely to be found.
- The **electron configuration** is the arrangement of electrons in an atom, molecule or other physical structure.
- Energy diagrams such as **Aufbau diagrams** are used to show the electron configuration of atoms.
- The electron configuration of an atom can be given using spectroscopic notation.
- Different orbitals have different shapes: s orbitals are spherically shaped and p orbitals are dumbbell shaped.
- The electrons in the outermost energy level are called **valence electrons**.
- The electrons in an atom that are not valence electrons are called **core electrons**.
- Atoms whose outermost energy level is full, are less chemically reactive and therefore more stable, than those atoms whose outermost energy level is not full.

Chapter 4 End of chapter exercises

1. Write down only the word/term for each of the following descriptions.
   a. The sum of the number of protons and neutrons in an atom
   b. The defined space around an atom's nucleus, where an electron is most likely to be found
2. For each of the following, say whether the statement is true or false. If it is false, re-write the statement correctly.
   a. $^{20}_{10}Ne$ and $^{22}_{10}Ne$ each have 10 protons, 12 electrons and 12 neutrons.
   b. The atomic mass of any atom of a particular element is always the same.
   c. It is safer to use helium gas rather than hydrogen gas in balloons.
   d. Group 1 elements readily form negative ions.
3. The three basic components of an atom are:
   a. protons, neutrons, and ions
   b. protons, neutrons, and electrons
   c. protons, neutrinos, and ions
   d. protium, deuterium, and tritium
4. The charge of an atom is:
   a. positive

   b. neutral
   c. negative
   d. none of the above

5. If Rutherford had used neutrons instead of alpha particles in his scattering experiment, the neutrons would:
   a. not deflect because they have no charge
   b. have deflected more often
   c. have been attracted to the nucleus easily
   d. have given the same results
6. Consider the isotope $^{234}_{92}$U. Which of the following statements is *true*?
   a. The element is an isotope of $^{234}_{94}$Pu
   b. The element contains 234 neutrons
   c. The element has the same electron configuration as $^{238}_{92}$U
   d. The element has an atomic mass number of 92
7. The electron configuration of an atom of chlorine can be represented using the following notation:
   a. $1s^2 2s^8 3s^7$
   b. $1s^2 2s^2 2p^6 3s^2 3p^5$
   c. $1s^2 2s^2 2p^6 3s^2 3p^6$
   d. $1s^2 2s^2 2p^5$
8. Give the standard notation for the following elements:
   a. beryllium
   b. carbon-12
   c. titanium-48
   d. fluorine
9. Give the electron configurations and Aufbau diagrams for the following elements:
   a. aluminium
   b. phosphorus
   c. carbon
   d. oxygen ion
   e. calcium ion
10. For each of the following elements give the number of protons, neutrons and electrons in the element:
    a. $^{195}_{78}$Pt
    b. $^{40}_{18}$Ar
    c. $^{59}_{27}$Co
    d. $^{7}_{3}$Li
    e. $^{11}_{5}$B
11. For each of the following elements give the element or number represented by x:
    a. $^{103}_{45}$X
    b. $^{35}_{x}$Cl

c. ${}^{x}_{4}Be$

12. Which of the following are isotopes of ${}^{24}_{12}Mg$:

a. ${}^{12}_{25}Mg$

b. ${}^{26}_{12}Mg$

c. ${}^{24}_{13}Al$

13. If a sample contains 69% of copper-63 and 31% of copper-65, calculate the relative atomic mass of an atom in that sample.

14. Complete the following table:

Element or ion	Electron configuration	Core electrons	Valence electrons
Boron (B)			
Calcium (Ca)			
Silicon (Si)			
Lithium ion (Li^{+})			
Neon (Ne)			

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(1.) 0019 (2.) 001a (3.) 001b (4.) 001c (5.) 001d (6.) 001e

(7.) 001f (8.) 001g (9.) 001h (10.) 001i (11.) 001j (12.) 001k

(13.) 001m (14.) 001n

The periodic table 5

The arrangement of the elements ESABM

The **periodic table of the elements** is a method of showing the chemical elements in a table with the elements arranged in order of increasing atomic number. Most of the work that was done to arrive at the periodic table that we know can be attributed to a Russian chemist named **Dmitri Mendeleev**. Mendeleev designed the table in 1869 in such a way that recurring ("periodic") trends or patterns in the properties of the elements could be shown. Using the trends he observed, he left gaps for those elements that he thought were "missing". He also predicted the properties that he thought the missing elements would have when they were discovered. Many of these elements were indeed discovered and Mendeleev's predictions were proved to be correct.
See introductory video: (Video: VParg at www.everythingscience.co.za)

To show the recurring properties that he had observed, Mendeleev began new rows in his table so that elements with similar properties were in the same vertical columns, called **groups**. Each row was referred to as a **period**. Figure 5.3 shows a simplified version of the periodic table. The full periodic table is reproduced at the front of this book. You can view an online periodic table at *http://periodictable.com/*.

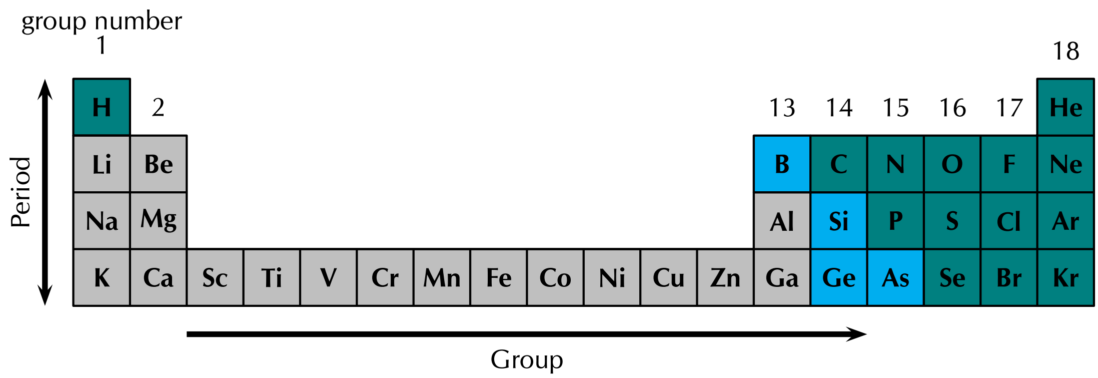


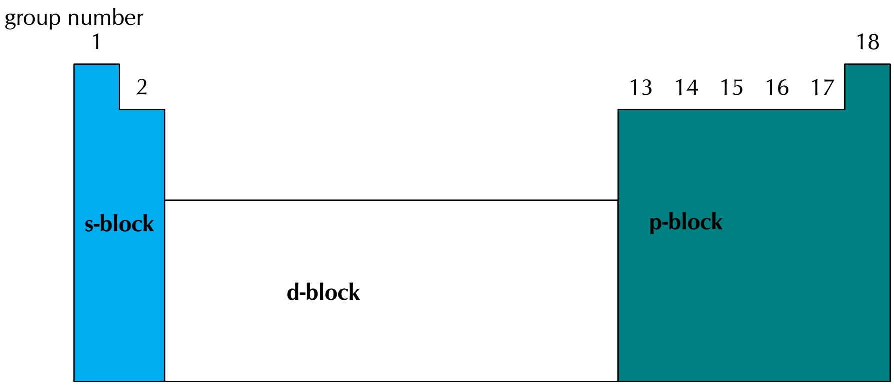
Figure 5.1: A simplified diagram showing part of the periodic table. Metals are given in gray, metalloids in light blue and non-metals in turquoise.

Definitions and important concepts ESABN

Before we can talk about the trends in the periodic table, we first need to define some terms that are used:

- **Atomic radius**
  The atomic radius is a measure of the size of an atom.
- **Ionisation energy**
  The first ionisation energy is the energy needed to remove one electron from an atom in the gas phase. The ionisation energy is different for each element. We can also define second, third, fourth, etc. ionisation energies. These are the energies needed to remove the second, third, or fourth electron respectively.
- **Electron affinity**
  Electron affinity can be thought of as how much an element wants electrons.
- **Electronegativity**
  Electronegativity is the tendency of atoms to attract electrons. The electronegativity of the elements starts from about 0.7 (Francium (Fr)) and goes up to 4 (Fluorine (F))
- A **group** is a vertical column in the periodic table and is considered to be the most important way of classifying the elements. If you look at a periodic table, you will see the groups numbered at the top of each column. The groups are numbered from left to right starting with 1 and ending with 18. This is the convention that we will use in this book. On some periodic tables you may see that the groups are numbered from left to right as follows: 1, 2, then an open space which contains the **transition elements**, followed by groups 3 to 8. Another way to label the groups is using Roman numerals.
- A **period** is a horizontal row in the periodic table of the elements. The periods are labelled from top to bottom, starting with 1 and ending with 7.

For each element on the periodic table we can give its period number and its group number. For example, B is in period 2 and group 13. We can also determine the electronic structure of an element from its position on the periodic table. In chapter 4 you worked out the electronic configuration of various elements. Using the periodic table we can easily give the electronic configurations of any element. To see how this works look at the following:



We also note that the period number gives the energy level that is being filled. For example, phosphorus (P) is in the third period and group 15. Looking at the figure above, we see that the p-orbital is being filled. Also the third energy level is being filled. So its electron configuration is: [Ne]$3s^2 3p^3$. (Phosphorus is in the third group in the p-block, so it must have 3 electrons in the p shell.)

Periods in the periodic table ESABO

The following diagram illustrates some of the key trends in the periods:

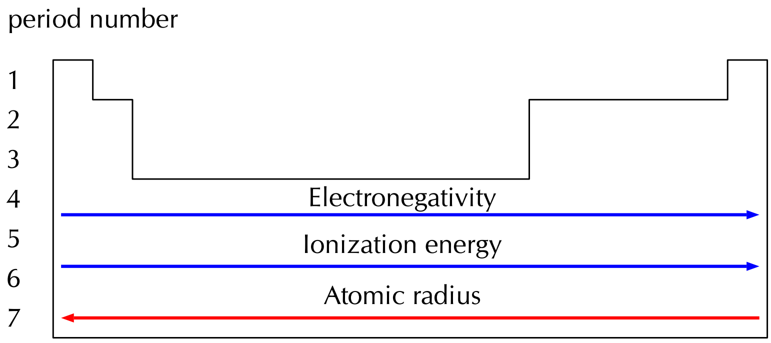


Figure 5.2: Trends on the periodic table.

See video: VParr at www.everythingscience.co.za

Table 5.1 summarises the patterns or trends in the properties of the elements in period 3. Similar trends are observed in the other periods of the periodic table. The chlorides are compounds with chlorine and the oxides are compounds with oxygen.

Element	$^{23}_{11}$Na	$^{24}_{12}$Mg	$^{27}_{13}$Al	$^{28}_{14}$Si	$^{31}_{15}$P	$^{32}_{16}$S	$^{35}_{17}$Cl
Chlorides	NaCl	$MgCl_2$	$AlCl_3$	$SiCl_4$	PCl_5 or PCl_3	S_2Cl_2	no chlorides
Oxides	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_6 or P_4O_{10}	SO_3 or SO_4	Cl_2O_7 or Cl_2O
Valence electrons	$3s^1$	$3s^2$	$3s^23p^1$	$3s^23p^2$	$3s^23p^3$	$3s^23p^4$	$3s^23p^5$
Atomic radius	Decreases across a period.						
First Ionization energy	The general trend is an increase across the period.						
Electronegativity	Increases across the period.						
Melting and boiling point	Increases to silicon and then decreases to argon.						
Electrical conductivity	Increases from sodium to aluminium. Silicon is a semi-conductor. The rest are insulators.						

Table 5.1: Summary of the trends in period 3

Note that we have left argon ($^{40}_{18}$Ar) out. Argon is a noble gas with electron configuration: [Ne]$3s^23p^6$. Argon does not form any compounds with oxygen or chlorine.

Exercise 5 - 1

1. Use Table 5.1 and Figure 5.2 to help you produce a similar table for the elements in period 2.
2. Refer to the data table below which gives the ionisation energy (in kJ $\cdot$ mol^{-1}) and atomic number (Z) for a number of elements in the periodic table:

Z	Name of element	Ionization energy	Z	Name of element	Ionization energy
1		1310	10		2072
2		2360	11		494
3		517	12		734
4		895	13		575
5		797	14		783
6		1087	15		1051
7		1397	16		994
8		1307	17		1250
9		1673	18		1540

a. Fill in the names of the elements.

b. Draw a line graph to show the relationship between atomic number (on the x-axis) and ionisation energy (y-axis).

c. Describe any trends that you observe.

d. Explain why:

- **i**. the ionisation energy for $Z = 2$ is higher than for $Z = 1$
- **ii**. the ionisation energy for $Z = 3$ is lower than for $Z = 2$
- **iii**. the ionisation energy increases between $Z = 5$ and $Z = 7$

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(1.) 001p (2.) 001q

Chemical properties of the groups

ESABP

In some groups, the elements display very similar chemical properties and some of the groups are even given special names to identify them. The characteristics of each group are mostly determined by the electron configuration of the atoms of the elements in the group. The names of the groups are summarised in Figure 5.3

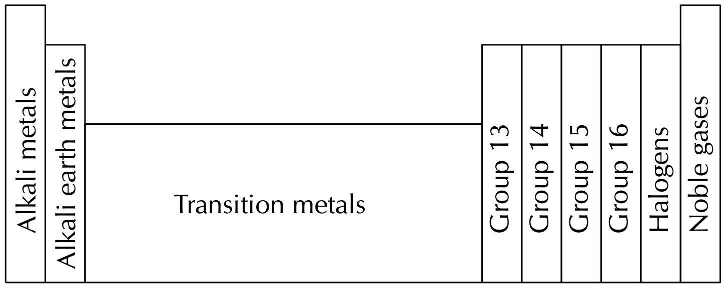


Figure 5.3: Groups on the periodic table

A few points to note about the groups are:

- Although hydrogen appears in group 1, it is not an alkali metal.
- Group 15 elements are sometimes called the pnictogens.
- Group 16 elements are sometimes known as the chalcogens.
- The **halogens** and the **alkali earth metals** are very reactive groups.
- The **noble gases** are *inert* (unreactive).

See video: VPasj at www.everythingscience.co.za

The following diagram illustrates some of the key trends in the groups of the periodic table:

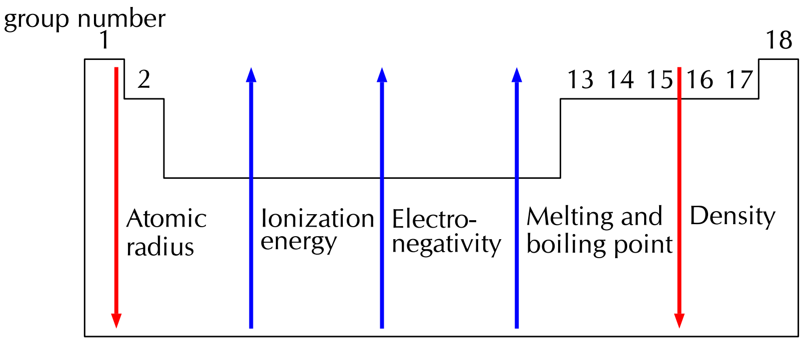


Figure 5.4: Trends in the groups on the periodic table.

Table 5.2 summarises the patterns or trends in the properties of the elements in group 1. Similar trends are observed for the elements in the other groups of the periodic table. We can use the information in 5.2 to predict the chemical properties of unfamiliar elements. For example, given the element Francium (Fr) we can say that its electronic structure will be [Rn]$7s^1$, it will have a lower first ionisation energy than caesium (Cs) and its melting and boiling point will also be lower than caesium.

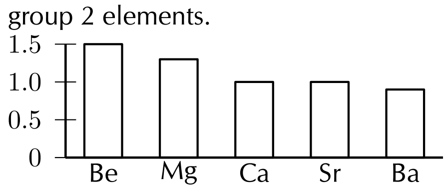
You should also recall from chapter 2 that the metals are found on the left of the periodic table, non-metals are on the right and metalloids are found on the zig-zag line that starts at boron.

Element	$^{7}_{3}$Li	$^{7}_{3}$Na	$^{7}_{3}$K	$^{7}_{3}$Rb	$^{7}_{3}$Cs
Electron structure	[He]$2s^1$	[Ne]$3s^1$	[Ar]$4s^1$	[Kr]$4s^1$	[Xe]$5s^1$
Group 1 chlorides	LiCl	NaCl	KCl	RbCl	CsCl
	Group 1 elements all form halogen compounds in a 1:1 ratio				
Group 1 oxides	Li_2O	Na_2O	K_2O	Rb_2O	Cs_2O
	Group 1 elements all form oxides in a 2:1 ratio				
Atomic radius	Increases as you move down the group.				
First ionisation energy	Decreases as you move down the group.				
Electronegativity	Decreases as you move down the group.				
Melting and boiling point	Decreases as you move down the group.				
Density	Increases as you move down the group.				

Table 5.2: Summary of the trends in group 1

Exercise 5 - 2

1. Use Table 5.2 and Figure 5.4 to help you produce similar tables for group 2 and group 17.
2. The following two elements are given. Compare these elements in terms of the following properties. Explain the differences in each case. $^{24}_{12}$Mg and $^{40}_{20}$Ca.
   a. Size of the atom (atomic radius)
   b. Electronegativity
   c. First ionisation energy
   d. Boiling point
3. Study the following graph and explain the trend in electronegativity of the group 2 elements.



4. Refer to the elements listed below:

- Lithium (Li)
- Chlorine (Cl)
- Magnesium (Mg)
- Neon (Ne)
- Oxygen (O)
- Calcium (Ca)
- Carbon (C)

Which of the elements listed above:

a. belongs to Group 1
b. is a halogen
c. is a noble gas
d. is an alkali metal
e. has an atomic number of 12
f. has four neutrons in the nucleus of its atoms
g. contains electrons in the 4th energy level
h. has all its energy orbitals full
i. will have chemical properties that are most similar

More practice video solutions or help at www.everythingscience.co.za

(1.) 001r (2.) 001s (3.) 001t (4.) 001u

Activity: *Inventing your own periodic table*

You are the official chemist for the planet Zog. You have discovered all the same elements that we have here on Earth, but you don't have a periodic table. The citizens of Zog want to know how all these elements relate to each other. How would you invent the periodic table? Think about how you would organise the data that you have and what properties you would include. Do not simply copy Mendeleev's ideas, be creative and come up with some of your own. Research other forms of the periodic table and make one that makes sense to you. Present your ideas to your class.

Circular periodic table

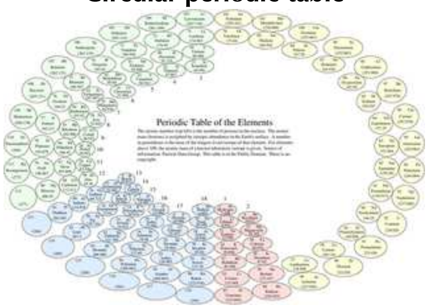


Image from Wikimedia commons

Chapter 5 | Summary

See the summary presentation (▶ Presentation: VPddg at www.everythingscience.co.za)

- Elements are arranged in periods and groups on the periodic table. The elements are arranged according to increasing atomic number.
- A **group** is a column on the periodic table containing elements with similar properties. A **period** is a row on the periodic table.
- The atomic radius is a measure of the size of the atom.
- The first ionisation energy is the energy needed to remove one electron from an atom in the gas phase.
- Electronegativity is the tendency of atoms to attract electrons.
- Across a period the ionisation energy and electronegativity increase. The atomic radius decreases across a period.
- The groups on the periodic table are labelled from 1 to 18. Group 1 is known as the alkali metals, group 2 is known as the alkali earth metals, group 17 is known as the halogens and the group 18 is known as the noble gases. The elements in a group have similar properties.
- The atomic radius and the density both increase down a group. The ionisation energy, electronegativity, and melting and boiling points all decrease down a group.

Chapter 5 End of chapter exercises

1. For the following questions state whether they are true or false. If they are false, correct the statement
   a. The group 1 elements are sometimes known as the alkali earth metals.
   b. The group 8 elements are known as the noble gases.
   c. Group 7 elements are very unreactive.
   d. The transition elements are found between groups 3 and 4.
2. Give one word or term for each of the following:
   a. The energy that is needed to remove one electron from an atom
   b. A horizontal row on the periodic table
   c. A very reactive group of elements that is missing just one electron from their outer shells.
3. Given $^{80}_{35}\text{Br}$ and $^{35}_{17}\text{Cl}$. Compare these elements in terms of the following properties. Explain the differences in each case.
   a. Atomic radius
   b. Electronegativity

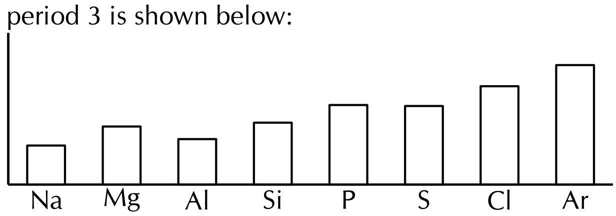
   c. First ionisation energy
   d. Boiling point

4. Given the following table:

Element	Na	Mg	Al	Si	P	S	Cl	Ar
Atomic number	11	12	13	14	15	16	17	18
Density ($g \cdot cm^{-3}$)	0,97	1,74	2,70	2,33	1,82	2,08	3,17	1,78
Melting point ($°C$)	370,9	923,0	933,5	1687	317,3	388,4	171,6	83,8
Boiling point ($°C$)	1156	1363	2792	3538	550	717,8	239,1	87,3
Electronegativity	0.93	1.31	1.61	1.90	2.19	2.58	3.16	-

   Draw graphs to show the patterns in the following physical properties:
   a. Density
   b. Boiling point
   c. Melting point
   d. Electronegativity

5. A graph showing the pattern in first ionisation energy for the elements in period 3 is shown below:



   a. Explain the pattern by referring to the electron configuration
   b. Predict the pattern in the first ionisation energy for the elements in period 2.
   c. Draw a rough graph to show the pattern predicted in the previous question.

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(1.) 001v (2.) 001w (3.) 001x (4.) 001y (5.) 001z

Chemical bonding 6

Introduction ESABQ

When you look at everything around you and what it is made of, you will realise that atoms seldom exist on their own. More often, the things around us are made up of different atoms that have been joined together. This is called **chemical bonding**. Chemical bonding is one of the most important processes in chemistry because it allows all sorts of different molecules and combinations of atoms to form, which then make up the objects in the complex world around us.

See introductory video: (Video: VPawb at www.everythingscience.co.za)

What happens when atoms bond? ESABR

A **chemical bond** is formed when atoms are held together by attractive forces. This attraction occurs when electrons are *shared* between atoms, or when electrons are *exchanged* between the atoms that are involved in the bond. The sharing or exchange of electrons takes place so that the outer energy levels of the atoms involved are filled, making the atoms are more stable. If an electron is **shared**, it means that it will spend its time moving in the electron orbitals around *both* atoms. If an electron is **exchanged** it means that it is transferred from one atom to another. In other words one atom *gains* an electron while the other *loses* an electron.

> ### DEFINITION: Chemical bond
>
> A chemical bond is the physical process that causes atoms and molecules to be attracted to each other and held together in more stable chemical compounds.

The type of bond that is formed depends on the elements that are involved. In this chapter, we will be looking at three types of chemical bonding: **covalent**, **ionic** and **metallic**

bonding.

You need to remember that it is the *valence electrons* (those in the outermost level) that are involved in bonding and that atoms will try to fill their outer energy levels so that they are more stable. The noble gases have completely full outer energy levels, so are very stable and do not react easily with other atoms.

Lewis structures

ESABS

Tip

If we write the condensed electron configuration, then we can easily see the valence electrons.

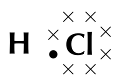
Lewis notation uses dots and crosses to represent the **valence electrons** on different atoms. The chemical symbol of the element is used to represent the nucleus and the inner electrons of the atom. To determine which are the valence electrons we look at the last *energy level* in the atom's electronic structure (chapter 4). For example, chlorine's electronic structure can be written as: $1s^22s^22p^63s^23p^5$ or $[Ne]3s^23p^5$. The last energy level is the third one and it contains 7 electrons. These are the valence electrons.

For example:

A hydrogen atom (one valence electron) would be represented like this: **H •**

A chlorine atom (seven valence electrons) would look like this: 

A molecule of hydrogen chloride would be shown like this:



The dot and cross in between the two atoms, represent the pair of electrons that are shared in the covalent bond.

Table 6.1 gives some further examples of Lewis diagrams.

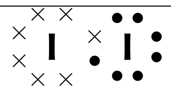
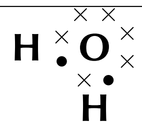
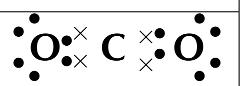
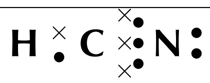
Iodine	I_2	
Water	H_2O	
Carbon dioxide	CO_2	
Hydrogen cyanide	HCN	

Table 6.1: Lewis diagrams for some simple molecules

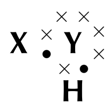
For carbon dioxide, you can see how we represent a double bond in Lewis notation. As there are two bonds between each oxygen atom and the carbon atom, two pairs of valence electrons link them. Similarly, hydrogen cyanide shows how to represent a triple bond.

Exercise 6 - 1

1. Represent each of the following *atoms* using Lewis notation:
   a. beryllium
   b. calcium
   c. lithium
2. Represent each of the following *molecules* using Lewis notation:
   a. bromine gas (Br_2)
   b. carbon dioxide (CO_2)

   Which of these two molecules contains a double bond?
3. Two chemical reactions are described below.
   - nitrogen and hydrogen react to form NH_3
   - carbon and hydrogen bond to form a molecule of CH_4

   For each reaction, give:
   a. the number of electrons in the outermost energy level
   b. the Lewis structure of the product that is formed
   c. the name of the product
4. A chemical compound has the following Lewis notation:



   a. How many valence electrons does element Y have?
   b. What is the valency of element Y? (remember that the valency of an atom is the number of chemical bonds it can form.)
   c. What is the valency of element X?
   d. How many covalent bonds are in the molecule?
   e. Suggest a name for the elements X and Y.

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(1.) 0020 (2.) 0021 (3.) 0022 (4.) 0023

Covalent Bonding

ESABT

The nature of the covalent bond

ESABU

Covalent bonding occurs between the atoms of **non-metals**. The outermost orbitals of the atoms overlap so that unpaired electrons in each of the bonding atoms can be shared. By overlapping orbitals, the outer energy shells of all the bonding atoms are filled. The shared electrons move in the orbitals around *both* atoms. As they move, there is an attraction between these negatively charged electrons and the positively charged nuclei. This attractive force holds the atoms together in a covalent bond.

See video: VPaxf at www.everythingscience.co.za

Tip

There is a relationship between the valency of an element and its position on the periodic table. For the elements in groups 1 and 2, the valency is the group number. For the elements in groups 13 - 18, the valency is the group number minus 10. For the transition metals, the valency can vary. In these cases we indicate the valency by a roman numeral after the element name, e.g. iron (III) chloride.

DEFINITION: Covalent bond

Covalent bonding is a form of chemical bonding where pairs of electrons are shared between atoms.

You will have noticed in table 6.1 that the number of electrons that are involved in bonding varies between atoms. We can say the following:

- A **single covalent** bond is formed when two electrons are shared between the same two atoms, one electron from each atom.
- A **double covalent** bond is formed when four electrons are shared between the same two atoms, two electrons from each atom.
- A **triple covalent** bond is formed when six electrons are shared between the same two atoms, three electrons from each atom.

You should also have noticed that compounds can have a mixture of single, double and triple bonds and that an atom can have several bonds. In other words, an atom does not need to share all its valence electrons with one other atom, but can share its valence electrons with several different atoms.
We say that the **valency** of the atoms is different.

DEFINITION: Valency

The number of electrons in the outer shell of an atom which are able to be used to form bonds with other atoms.

Below are a few examples. Remember that it is only the *valence electrons* that are involved in bonding and so when diagrams are drawn to show what is happening during bonding, it is only these electrons that are shown.

Example 1: *Covalent bonding*

QUESTION

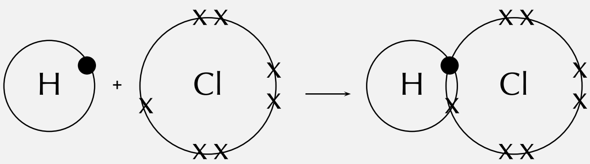
How do hydrogen and chlorine atoms bond covalently in a molecule of hydrogen chloride?

SOLUTION

Step 1 : ***Determine the electron configuration of each of the bonding atoms.***
A chlorine atom has 17 electrons and an electron configuration of $[Ne]3s^{2}3p^{5}$. A hydrogen atom has only one electron and an electron configuration of $1s^{1}$.

Step 2 : ***Determine how many of the electrons are paired or unpaired.***
Chlorine has seven valence electrons. One of these electrons is unpaired. Hydrogen has one valence electron and it is unpaired.

Step 3 : ***Work out how the electrons are shared***
The hydrogen atom needs one more electron to complete its outermost energy level. The chlorine atom also needs one more electron to complete its outermost energy level. Therefore *one pair of electrons* must be shared between the two atoms. A single covalent bond will be formed.



Example 2: *Covalent bonding involving multiple bonds*

QUESTION

How do nitrogen and hydrogen atoms bond to form a molecule of ammonia (NH_3)?

SOLUTION

Step 1 : ***Give the electron configuration***

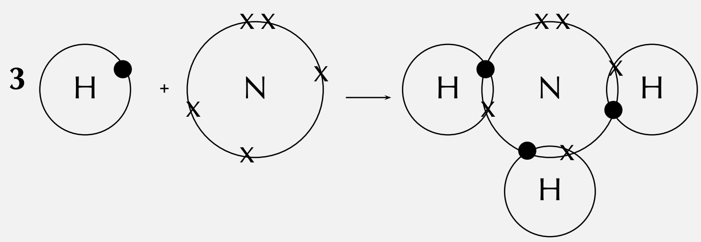
A nitrogen atom has seven electrons, and an electron configuration of [He]$2s^2 2p^3$. A hydrogen atom has only one electron, and an electron configuration of $1s^1$.

Step 2 : ***Give the number of valence electrons***

Nitrogen has five valence electrons. Three of these electrons are unpaired. Hydrogen has one valence electron and it is unpaired.

Step 3 : ***Work out how the electrons are shared***

Each hydrogen atom needs one more electron to complete its valence energy shell. The nitrogen atom needs three more electrons to complete its valence energy shell. Therefore *three pairs of electrons* must be shared between the four atoms involved. Three single covalent bonds will be formed.



Example 3: *Covalent bonding involving a double bond*

QUESTION

How do oxygen atoms bond covalently to form an oxygen molecule?

SOLUTION

Step 1 : ***Determine the electron configuration of the bonding atoms.***

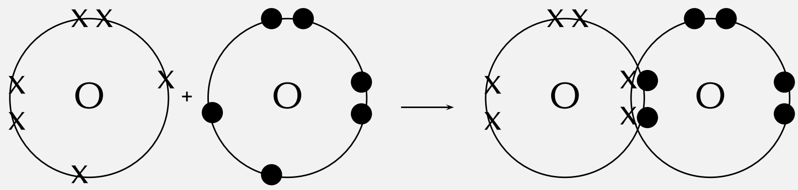
Each oxygen atom has eight electrons, and their electron configuration is $[He]2s^22p^4$.

Step 2 : ***Determine the number of valence electrons for each atom and how many of these electrons are paired and unpaired.***

Each oxygen atom has six valence electrons. Each atom has two unpaired electrons.

Step 3 : ***Work out how the electrons are shared***

Each oxygen atom needs two more electrons to complete its valence energy shell. Therefore *two pairs of electrons* must be shared between the two oxygen atoms so that both outermost energy levels are full. A double bond is formed.



Properties of covalent compounds

ESABV

Covalent compounds have several properties that distinguish them from ionic compounds and metals. These properties are:

1. The melting and boiling points of covalent compounds are generally lower than those of ionic compounds.
2. Covalent compounds are generally more flexible than ionic compounds. The molecules in covalent compounds are able to move around to some extent and can sometimes slide over each other (as is the case with graphite, which is why the lead in your pencil feels slightly slippery). In ionic compounds, all the ions are tightly held in place.
3. Covalent compounds generally are not very soluble in water, for example plastics are covalent compounds and many plastics are water resistant.
4. Covalent compounds generally do not conduct electricity when dissolved in water, for example iodine dissolved in pure water does not conduct electricity.

Exercise 6 - 2

1. Explain the difference between the *valence electrons* and the *valency* of an element.

2. Complete the table below by filling in the number of valence electrons for each of the elements shown:

Element	**Group number**	**No. of valence electrons**	**No. of electrons needed to fill outer shell**
He			
Li			
B			
C			
F			
Ne			
Na			
Al			
P			
S			
Ca			
Kr			

3. Draw simple diagrams to show how electrons are arranged in the following covalent molecules:
   a. hydrogen sulphide (H_2S)
   b. chlorine (Cl_2)
   c. nitrogen (N_2)
   d. carbon monoxide (CO)

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(1.) 0025 (2.) 0026 (3.) 0027

Ionic bonding ESABW

The nature of the ionic bond ESABX

When electrons are *transferred* from one atom to another it is called **ionic bonding**.

Electronegativity is a property of an atom, describing how strongly it attracts or holds onto electrons. Ionic bonding takes place when the difference in electronegativity between the two atoms is more than 1.7. This usually happens when a metal atom bonds with a non-metal atom. When the difference in electronegativity is large, one atom will attract the shared electron pair much more strongly than the other, causing electrons to be transferred to the atom with higher electronegativity. When ionic bonds form, a metal donates one or more electrons, due to having a low electronegativity, to form a positive ion or cation. The non-metal atom has a high electronegativity, and therefore readily gains electrons to form a negative ion or anion. The two ions are then attracted to each other by electrostatic forces.

See video: VPaxl at www.everythingscience.co.za

DEFINITION: Ionic bond

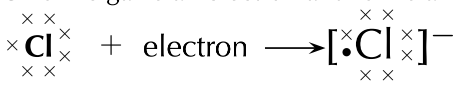
An ionic bond is a type of chemical bond where one or more electrons are transferred from one atom to another.

Example 1:

In the case of NaCl, the difference in electronegativity between Na (0,93) and Cl (3,16) is $2,1$. Sodium has only one valence electron, while chlorine has seven. Because the electronegativity of chlorine is higher than the electronegativity of sodium, chlorine will attract the valence electron of the sodium atom very strongly. This electron from sodium is transferred to chlorine. Sodium loses an electron and forms an Na^{+} ion.

$$\mathbf{Na}\bullet \longrightarrow \mathbf{Na}^{+} + \text{electron}$$

Chlorine gains an electron and forms an Cl^{-} ion.



The electron is therefore transferred from sodium to chlorine:

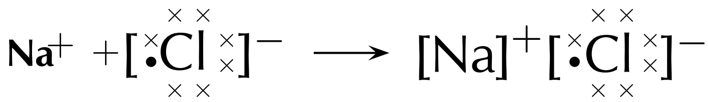


Figure 6.2: Ionic bonding in sodium chloride

The balanced equation for the reaction is:

$$2Na + Cl_2 \rightarrow 2NaCl$$

Example 2:

Another example of ionic bonding takes place between magnesium (Mg) and oxygen (O_2) to form magnesium oxide (MgO). Magnesium has two valence electrons and an electronegativity of $1,31$, while oxygen has six valence electrons and an electronegativity of $3,44$. Since oxygen has a higher electronegativity, it attracts the two valence electrons from the magnesium atom and these electrons are transferred from the magnesium atom to the oxygen atom. Magnesium loses two electrons to form Mg^{2+}, and oxygen gains two electrons to form O^{2-}. The attractive force between the oppositely charged ions is what holds the compound together.

The balanced equation for the reaction is:

$$2Mg + O_2 \rightarrow 2MgO$$

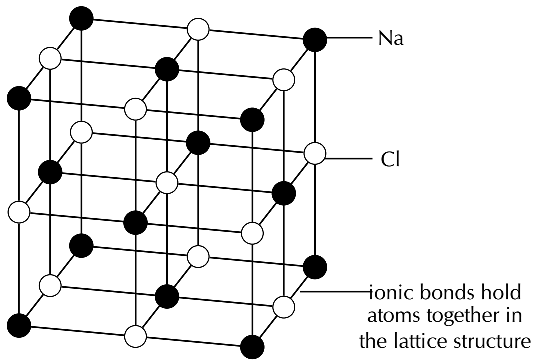
Because oxygen is a diatomic molecule, two magnesium atoms will be needed to combine with one oxygen molecule (which has two oxygen atoms) to produce two units of magnesium oxide (MgO).

Note

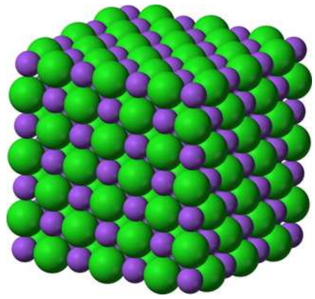
Chlorine is a diatomic molecule and so for it to take part in ionic bonding, it must first break up into two atoms of chlorine. Sodium is part of a metallic lattice and the individual atoms must first break away from the lattice.

The crystal lattice structure of ionic compounds ESABY

Ionic substances are actually a combination of lots of ions bonded together into a giant molecule. The arrangement of ions in a regular, geometric structure is called a **crystal lattice**. So in fact NaCl does not contain one Na and one Cl ion, but rather a lot of these two ions arranged in a crystal lattice where the ratio of Na to Cl ions is 1:1. The structure of the crystal lattice is shown below.



The crystal lattice arrangement in NaCl



A space filling model of the sodium chloride lattice

Properties of Ionic Compounds ESABZ

Ionic compounds have a number of properties:

1. Ions are arranged in a lattice structure
2. Ionic solids are crystalline at room temperature
3. The ionic bond is a strong electrostatic attraction. This means that ionic compounds are often hard and have high melting and boiling points
4. Ionic compounds are brittle and bonds are broken along planes when the compound is put under pressure (stressed)
5. Solid crystals do not conduct electricity, but ionic solutions do

Exercise 6 - 3

1. Explain the difference between a *covalent* and an *ionic* bond.

2. Magnesium and chlorine react to form magnesium chloride.
   a. What is the difference in electronegativity between these two elements?
   b. Give the chemical formula for:
      i. a magnesium ion
      ii. a chloride ion
      iii. the ionic compound that is produced during this reaction
   c. Write a balanced chemical equation for the reaction that takes place.
3. Draw Lewis diagrams to represent the following ionic compounds:
   a. sodium iodide (NaI)
   b. calcium bromide ($CaBr_2$)
   c. potassium chloride (KCl)

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(1.) 0028 (2.) 0029 (3.) 002a

Metallic Bonding ESACA

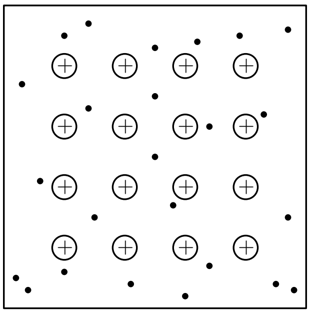
The nature of the metallic bond ESACB

The structure of a metallic bond is quite different from covalent and ionic bonds. In a metallic bond, the valence electrons are *delocalised*, meaning that an atom's electrons do not stay around that one nucleus. In a metallic bond, the positive atomic nuclei (sometimes called the "atomic kernels") are surrounded by a sea of delocalised electrons which are attracted to the nuclei (see figure below).

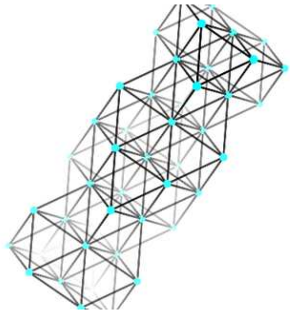
See video: VPaxw at www.everythingscience.co.za

DEFINITION: Metallic bond

Metallic bonding is the electrostatic attraction between the positively charged atomic nuclei of metal atoms and the delocalised electrons in the metal.



Positive atomic nuclei (+) surrounded by delocalised electrons (•)



Ball and stick model of copper

Properties of metals

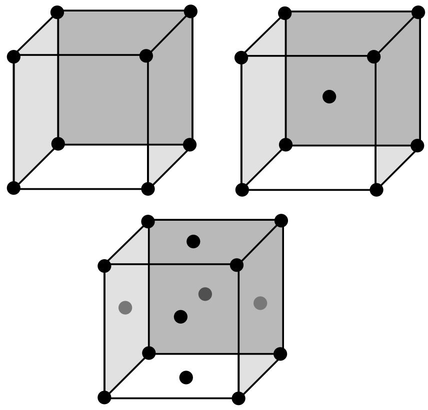
ESACC

1. Metals are *shiny*.
2. Metals *conduct electricity* because electrons are free to move.
3. Metals *conduct heat* because the positive nuclei are packed closely together and can easily transfer the heat.
4. Metals have a *high melting point* because the bonds are strong and a *high density* because of the tight packing of the nuclei.

Activity:	*Building models*

Using coloured balls (or jellytots) and sticks (or toothpicks) build models of each type of bonding. Think about how to represent each kind of bonding. For example, covalent bonding could be represented by simply connecting the balls with sticks to represent the molecules, while for ionic bonding you may wish to construct part of the crystal lattice.

Do some research on types of crystal lattices (although the section on ionic bonding only showed the crystal lattice for sodium chloride, many other types of lattices exist) and try to build some of these. Share your findings with your class and compare notes to see what types of crystal lattices they found. How would you show metallic bonding?



Exercise 6 - 4

1. Give two examples of everyday objects that contain:
   a. covalent bonds
   b. ionic bonds
   c. metallic bonds
2. Complete the table which compares the different types of bonding:

	Covalent	**Ionic**	**Metallic**
Types of atoms involved			
Nature of bond between atoms			
Melting point (high/low)			
Conducts electricity? (yes/no)			
Other properties			

3. Complete the table below by identifying the type of bond (covalent, ionic or metallic) in each of the compounds:

Molecular formula	**Type of bond**
H_2SO_4	
FeS	
NaI	
$MgCl_2$	
Zn	

4. Use your knowledge of the different types of bonding to explain the following statements:

a. A sodium chloride crystal does not conduct electricity.
b. Most jewellery items are made from metals.
c. It is very hard to break a diamond.
d. Pots are made from metals, but their handles are made from plastic.

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(1.) 002b (2.) 002c (3.) 002d (4.) 002e

Writing formulae

ESACD

In chapter 2 you learnt about the writing of chemical formulae. Table 6.2 shows some of the common anions and cations that you should know.

Name of compound ion	formula	Name of compound ion	formula
Acetate (ethanoate)	CH_3COO^-	Manganate	MnO_4^{2-}
Ammonium	NH_4^+	Nitrate	NO_3^-
Carbonate	CO_3^{2-}	Nitrite	NO_2^-
Chlorate	ClO_3^-	Oxalate	$C_2O_4^{2-}$
Chromate	CrO_4^{2-}	Oxide	O^{2-}
Cyanide	CN^-	Permanganate	MnO_4^-
Dihydrogen phosphate	$H_2PO_4^-$	Peroxide	O_2^{2-}
Hydrogen carbonate	HCO_3^-	Phosphate	PO_4^{3-}
Hydrogen phosphate	HPO_4^{3-}	Phosphide	P^{3-}
Hydrogen sulphate	HSO_4^-	Sulphate	SO_4^{2-}
Hydrogen sulphite	HSO_3^-	Sulphide	S^{2-}
Hydroxide	OH^-	Sulphite	SO_3^{2-}
Hypochlorite	ClO^-	Thiosulphate	$S_2O_3^{2-}$

Table 6.2: Table showing common compound ions and their formulae

Chemical compounds: names and masses ESACE

In chapter 4 you learnt about atomic masses. In this chapter we have learnt that atoms can combine to form compounds. Molecules are formed when atoms combine through covalent bonding, for example ammonia is a molecule made up of three hydrogen atoms and one nitrogen atom. The **relative molecular mass** (M) of ammonia (NH_3) is:

$$\begin{aligned} M &= \text{relative atomic mass of one nitrogen} + \text{relative atomic mass of three hydrogens} \\ &= 14,0 + 3(1,01) \\ &= 17,03 \end{aligned}$$

One molecule of NH_3 will have a mass of $17,03$ units. When sodium reacts with chlorine to form sodium chloride, we do not get a molecule of sodium chloride, but rather a sodium chloride crystal lattice. Remember that in ionic bonding molecules are not formed. We can also calculate the mass of one unit of such a crystal. We call this a **formula unit** and the mass is called the **formula mass**. The formula mass for sodium chloride is:

$$\begin{aligned} M &= \text{relative atomic mass of one sodium atom} + \text{relative atomic mass of one chlorine atom} \\ &= 23,0 + 35,45 \\ &= 58,45 \end{aligned}$$

The formula mass for NaCl is $58,45$ units.

Exercise 6 - 5

1. Write the chemical formulae for each of the following compounds and calculate the relative molecular mass or formula mass:
   a. hydrogen cyanide
   b. carbon dioxide
   c. sodium carbonate
   d. ammonium hydroxide
   e. barium sulphate
   f. copper (II) nitrate
2. Complete the following table. The cations at the top combine with the anions on the left. The first row is done for you. Also include the names of the compounds formed and the anions.

	Na^{+}	**Mg^{2+}**	**Al^{3+}**	**NH_4^{+}**	**H^{+}**
Br^{-} **name:**	NaBr sodium bromide	$MgBr_2$ magnesium bromide	$AlBr_3$ aluminium bromide	$(NH_4)Br$ ammonium bromide	HBr hydrogen bromide
S^{2-} **name:**					
P^{3-} **name:**					
MnO_4^{-} **name:**					
$Cr_2O_7^{2-}$ **name:**					
HPO_4^{2-} **name:**					

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(1.) 002f (2.) 002g

Chapter 6 | Summary

See the summary presentation (Presentation: VPduz at www.everythingscience.co.za)

- A **chemical bond** is the physical process that causes atoms and molecules to be attracted to each other and held together in more stable chemical compounds.
- Atoms are more **reactive**, and therefore more likely to bond, when their outer electron orbitals are not full. Atoms are less reactive when these outer orbitals contain the maximum number of electrons. This explains why the noble gases do not react.
- **Lewis** notation is one way of representing molecular structure. In Lewis notation, dots and crosses are used to represent the valence electrons around the central atom.
- When atoms bond, electrons are either shared or exchanged.

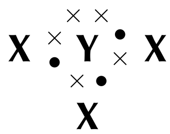
- **Covalent bonding** occurs between the atoms of non-metals and involves a sharing of electrons so that the orbitals of the outermost energy levels in the atoms are filled.
- A **double** or **triple bond** occurs if there are two or three electron pairs that are shared between the same two atoms.
- The **valency** is the number of electrons in the outer shell of an atom which are able to be used to form bonds with other atoms.
- Covalent compounds have lower melting and boiling points than ionic compounds. Covalent compounds are also generally flexible, are generally not soluble in water and do not conduct electricity.
- An **ionic bond** occurs between atoms where there is a large difference in electronegativity. An exchange of electrons takes place and the atoms are held together by the electrostatic force of attraction between the resulting oppositely-charged ions.
- Ionic solids are arranged in a **crystal lattice** structure.
- Ionic compounds have high melting and boiling points, are brittle in nature, have a lattice structure and are able to conduct electricity when in solution.
- A **metallic bond** is the electrostatic attraction between the positively charged nuclei of metal atoms and the delocalised electrons in the metal.
- Metals are able to conduct heat and electricity, they have a metallic lustre (shine), they are both malleable (flexible) and ductile (stretchable) and they have a high melting point and density.
- We can work out the relative molecular mass for covalent compounds and the formula mass for ionic compounds and metals.

Chapter 6 End of chapter exercises

1. Explain the meaning of each of the following terms
   a. ionic bond
   b. covalent bond
2. Which ONE of the following best describes the bond formed between carbon and hydrogen?
   a. metallic bond
   b. covalent bond
   c. ionic bond
3. Which of the following reactions will *not* take place? Explain your answer.
   a. $H + H \rightarrow H_2$
   b. $Ne + Ne \rightarrow Ne_2$
   c. $Cl + Cl \rightarrow Cl_2$
4. Draw the Lewis structure for each of the following:

a. calcium
b. iodine
c. hydrogen bromide (HBr)
d. nitrogen dioxide (NO_2)

5. Given the following Lewis structure, where X and Y each represent a different element:



a. What is the valency of X?
b. What is the valency of Y?
c. Which elements could X and Y represent?

6. Complete the following table:

	K^+	Ca^{2+}	NH_4^+
OH^-			
O^{2-}			
NO_3^-			
PO_4^{3-}			

7. Potassium dichromate is dissolved in water.
a. Give the name and chemical formula for each of the ions in solution.

b. What is the chemical formula for potassium dichromate?

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(1.) 002h (2.) 002i (3.) 002j (4.) 002k (5.) 002m (6.) 002n
(7.) 002p